



Day 4 Live Session

00:00 - 00:05

Audio shared by Sumedha: City likes like diamonds blink on tight streets.

00:06 - 00:11

Audio shared by Sumedha: Honking horns play symphonies where chaos meets.

00:14 - 00:21

Audio shared by Sumedha: Unpainted stage In this moonlit traffic Let's turn another page

00:22 - 00:26

Audio shared by Sumedha: Morning coffee soup, and these rice, the dawn.

00:28 - 00:32

Audio shared by Sumedha: Busy bees in their carpool, just carry on.

00:33 - 00:35

Audio shared by Sumedha: rice, fair.

00:36 - 00:38

Audio shared by Sumedha: That is still surreal.

00:39 - 00:44

Audio shared by Sumedha: In the smooth trap, Our secrets, they conceal.

00:57 - 00:58

Audio shared by Sumedha: It's true.

01:01 - 01:04

Audio shared by Sumedha: Every riya story has to die.

01:06 - 01:10

Audio shared by Sumedha: Boys scream with faces we don't recognize.

01:12 - 01:15

Audio shared by Sumedha: Every mile within our hearts

01:22 - 01:27

Audio shared by Sumedha: In this moment, traffic, Freedom's what we find.

01:28 - 01:33

Audio shared by Sumedha: Silent radios play the songs of yesteryear.

01:33 - 01:35

Audio shared by Sumedha: Windows down the wind.

01:36 - 01:43

Audio shared by Sumedha: Arun Nau Tunnel vision through the lines of space and time.

02:55 - 02:58

Audio shared by Sumedha: Every life story has to fall

03:30 - 03:45

Audio shared by Sumedha: I walked these halls a thousand times, felt the silence right at the signs, hiding shadows, biting tongues, I thought I'd change outfit, I was used to it.

03:45 - 03:55

Audio shared by Sumedha: I tried to fit inside your world Bend myself a broken stead. Every whisper, every lie kept me hanging in this rope.

04:08 - 04:09

Audio shared by Sumedha: I just stayed too long.

04:19 - 04:28

Audio shared by Sumedha: I held my breath, I played along Felt the weight of staying strong
Every promise wore me thin Still I stayed and let you in.

04:29 - 04:31

Audio shared by Sumedha: I tried to fit inside your own.

04:32 - 04:35

Audio shared by Sumedha: And myself, the broken stead. Every whisper.

04:36 - 04:39

Audio shared by Sumedha: Kept me hanging in this robe.

04:46 - 04:47

Audio shared by Sumedha: The game soon.

04:50 - 04:51

Audio shared by Sumedha: below.

04:51 - 04:53

Audio shared by Sumedha: I'll just stayed too long

05:02 - 05:12

Audio shared by Sumedha: And now it's crystal clear. I was just a ghost to you. No more waiting, no more pain, I'm not afraid to walk away.

05:12 - 05:14

Audio shared by Sumedha: I just stayed too long

05:24 - 05:28

Audio shared by Sumedha: Feel... I feel like it, I...

05:35 - 05:37

Audio shared by Sumedha: I'll just stayed too long

05:54 - 05:55

Audio shared by Sumedha: How'd you stay tuned!

05:57 - 05:58

Audio shared by Sumedha: Now I'm gone.

05:59 - 06:01

Audio shared by Sumedha: I'll prove myself.

06:01 - 06:05

Audio shared by Sumedha: Try to fiddle along, I will find a place to belong.

07:05 - 07:06

Audio shared by Sumedha: Is this country?

07:07 - 07:10

Audio shared by Sumedha: Is this country? Is this country...

07:17 - 07:30

Audio shared by Sumedha: She's got surpass in happily drowned. Nair with frequently got around. Kapil and keep falling asleep.

07:30 - 07:43

Audio shared by Sumedha: We've gotta hear what she. And she's grace is it which you wasted Desai worries, that's all it takes is a change. I race, I give the

07:44 - 07:48

Audio shared by Sumedha: The best he comes, I give a card up.

07:50 - 07:58

Audio shared by Sumedha: He's not... Public... hunts happening It's got You falling back, no plans.

07:59 - 08:08

Audio shared by Sumedha: She has strength, she's gone in about an exceed at night. She likes to see our Joshi's

08:09 - 08:12

Audio shared by Sumedha: This song is gonna be bad.

08:16 - 08:20

Audio shared by Sumedha: She's out to where you want to.

08:25 - 08:28

Audio shared by Sumedha: some happy chill.

08:30 - 08:32

Audio shared by Sumedha: Get the sausage equipment.

08:58 - 09:01

Audio shared by Sumedha: She's not so pleasing habit.

09:01 - 09:06

Audio shared by Sumedha: May we speak well, he got a laugh.

09:06 - 09:07

Audio shared by Sumedha: cough.

09:07 - 09:11

Audio shared by Sumedha: The Jacqueline.

09:12 - 09:14

Audio shared by Sumedha: Gotta help us with.

09:14 - 09:18

Audio shared by Sumedha: And she's crazy to which you wasted Desai

09:20 - 09:24

Audio shared by Sumedha: The holy face gives a change.

09:27 - 09:29

Audio shared by Sumedha: Sounds like even now.

09:31 - 09:36

Audio shared by Sumedha: Is that... Holy hearts, I live.

09:37 - 09:45

Audio shared by Sumedha: You're falling back, no, She's got Let's bad.

09:48 - 09:50

Audio shared by Sumedha: Your judges!

10:11 - 10:14

Audio shared by Sumedha: sea of trashies.

10:14 - 10:17

Audio shared by Sumedha: Long is gonna be one.

10:23 - 10:25

Audio shared by Sumedha: zoo, zoo

10:49 - 10:54

Audio shared by Sumedha: City likes like diamonds blink on tight streets.

10:55 - 11:00

Audio shared by Sumedha: Honking horns play symphonies where chaos meets.

11:01 - 11:10

Audio shared by Sumedha: Like shadows on a neon painted stage. In this moonlit traffic, let's turn another page

11:11 - 11:15

Audio shared by Sumedha: Morning coffee symphonies rise with the dawn.

11:16 - 11:20

Audio shared by Sumedha: Busy bees in their carpool, just carry on.

11:22 - 11:27

Audio shared by Sumedha: Street lights fade, but dreams we've had are still surreal.

11:27 - 11:32

Audio shared by Sumedha: In this moment, Our secrets, they conceal.

11:41 - 11:42

Audio shared by Sumedha: Things be wrong.

11:46 - 11:47

Audio shared by Sumedha: This guy is true!

11:49 - 11:54

Audio shared by Sumedha: Every life story etched upon the world.

11:55 - 11:58

Audio shared by Sumedha: Boards scream with faces we don't recognize.

12:00 - 12:03

Audio shared by Sumedha: Every mile... within our...

12:08 - 12:10

Audio shared by Sumedha: We left behind.

12:11 - 12:16

Audio shared by Sumedha: In this moment, traffic... Freedom's what we find.

12:16 - 12:21

Audio shared by Sumedha: Silent radios play the songs of yesteryear.

12:22 - 12:23

Audio shared by Sumedha: Windows down the wind.

12:24 - 12:25

Audio shared by Sumedha: Arunaboo.

12:36 - 12:45

Shasti: Hello, everyone. Welcome back. Welcome back to day 4 of your AI Tenderlust Agalator program. How are you guys doing?

12:46 - 12:53

Shasti: Good morning, good evening, good afternoon, whichever part of the world you guys are joining from. How are you guys doing?

12:54 - 12:56

Shasti: Still okay? Too tired?

12:58 - 13:00

Shasti: Good evening, good morning!

13:02 - 13:05

Shasti: Doing great, great energy.

13:05 - 13:15

Shasti: Too exhausted. Folks, we are still in Day 4 of our program. I think too exhausted should not be the answer that we all should be getting now, because we still have a long way to go.

13:20 - 13:21

Shasti: Okay.

13:22 - 13:31

Shasti: I still see a couple of us have not yet joined the session, so we'll just wait for them for another minute or two, and then we will get started, right?

13:32 - 13:34

Shasti: In the meantime, folks.

13:34 - 13:47

Shasti: I have a very important question. As I said, we are in Day 4 of the program, and we are almost halfway through week 1 of our AI Ags Letter program. So, wanted to check, how are you guys doing so far?

13:47 - 14:03

Shasti: starting from day one, we have covered Level 1, Level 2, Level 3, and we got started with Level 4 yesterday, which is the world of automations, and we are gonna continue today with Level 4 as well. So, on a scale of 1 to 10, how are we doing?

14:06 - 14:24

Shasti: Better than day one, 10 exciting but overwhelming folks. As we keep saying, starting from our kickoff, it is going to be intense, and you will feel overwhelmed at one point, so do not worry about it. You have everything that you need, the recordings and the kind of support that you want, so make sure that, you are trying to keep up.

14:27 - 14:29

Shasti: Okay, interesting.

14:30 - 14:32

Shasti: Able to learn, good pace.

14:33 - 14:34

Shasti: Great for growth.

14:38 - 14:41

Shasti: Somebody says break time should be 10 minutes.

14:44 - 14:49

Shasti: Okay. Allow me a couple of seconds to go through all the messages that are in the chat.

14:50 - 14:53

Shasti: Learned a lot, learning a lot.

14:57 - 14:59

Shasti: Too much info.

14:59 - 15:19

Shasti: Yes, Praveen, completely agree, because as we said, the whole objective of the program is to, give all the possibilities of AI knowledge that is available out there, and if you anytime feel that it is too much to cope up with, take your time, you can go through the recordings at your own pace, do the workbooks later, and then put your questions on Ask Our channel.

15:21 - 15:29

Shasti: Neil Khan says my AI graph is upward at the moment. I think then we all are on the right page with the right objective that we are trying to achieve.

15:29 - 15:30

Shasti: Right?

15:30 - 15:33

Shasti: Really enjoying the hands-on? Perfect.

15:34 - 15:37

Shasti: Somebody says my brain overheats.

15:37 - 15:42

Shasti: I hope it gets cooled down by the time we end our program on day 14.

15:45 - 16:02

Shasti: Perfect! Thank you so much for all those messages. I think with that, we are clear that we are on the right path, and remember, we are just in Day 4 of the program. We still have

another 10 days to finish, so anytime you feel intense or overwhelmed, make sure that you're taking your time to go through the recordings if you're not gonna make it live, right?

16:03 - 16:05

Shasti: Perfect.

16:05 - 16:24

Shasti: Now, without any further ado, we will quickly get started with today's session. So, if you remember, yesterday we got introduced to the world of automations, and we built our first ever website lead spam checker using our Ann, right? How many of you actually, did the workflow either in breakout rooms or after the breakout rooms?

16:24 - 16:27

Shasti: Drop a me in the chat if you're able to successfully execute it.

16:30 - 16:35

Shasti: Okay, good to see 90% of you saying yes and me. Great.

16:37 - 16:51

Shasti: API key was disabled today at 6pm, folks. If you have tried to use the same API key that was in the workbook post 6PM ISD today, then you might face issues, because as we mentioned yesterday, API key was enabled till 6pm today.

16:54 - 16:55

Shasti: Great.

16:55 - 17:19

Shasti: So, we are gonna pick up from where we left yesterday. So yesterday, you got introduced to the world of automations, and you created your workflows. So today, what we are gonna do is, we are going to get introduced to the world of AI agents. Now, as Sumedha mentioned yesterday about the kind of use cases that you would want to automate, AI agents is where, you will learn about it, because now, let's say if I want to automate a process.

17:19 - 17:41

Shasti: that I know the SOP for. Now, maybe a Gmail sender. Every day I have to sit and read all my emails, and then I have to shoot Gmails or replies for the same. I have an SOP, but then I do not have the time to manually do it. That's where AI Agents comes into the picture. I can just go ahead and build an agent for it, and the agent will do the job for me. Like this, you can have

17:41 - 17:49

Shasti: multiple use cases of your own, maybe building a recruiter, recruitment agent, or an HR agent, or maybe anything that is

17:49 - 18:08

Shasti: applicable to your job and your role, right? That's when you can also automate most of the manual stuff that you're doing every day now, which is taking a lot of your time. So that's why we have planned this session today, where we are going to build a customer support agent using the AI Agent feature that is available out there in ATT.

18:08 - 18:23

Shasti: Now, if you ask me a question about, is the Ann is the only tool that we can make use to build an agent? No. There are multiple tools available out there. You have Zapier, you have MAG, you can build agents through codecs, through, even Clot, Clot Code, right?

18:23 - 18:34

Shasti: But why we are choosing NATN is because with the free version of NATEN, you can go ahead and create your agents, and also it is a widely used tool available out there. That's why we are choosing to NATON.

18:35 - 18:36

Shasti: I hope that is clear.

18:37 - 18:52

Shasti: I see a lot of questions about API keys, etc, etc, in the chat. Folks, please do put it on the Q&A box, as we discussed, starting from day one, Q&A box is where you can drop your questions, right? Please do go ahead and drop your questions on

18:52 - 19:01

Shasti: Q&A box. Where can I find the API key? API key was given as part of your workbook yesterday, and for today's workbook also, you'll get an API key.

19:03 - 19:27

Shasti: Great. So this is what we are going to do in today's session, as every day we will have two segments. One is where you will be learning, and second is where you will be doing the hands-on in the breakout rooms. So please make sure when the mentor is teaching, you're not going out there and doing it along with the mentor, because we have a dedicated breakout

room activity where you will be building your customer support Raj bot in the breakout rooms with the help of co-mentors.

19:27 - 19:37

Shasti: So please try not to follow along with the mentor when the mentor is trying to teach. And you'll also get introduced to a concept called RAT, which we have not learned in the last 3 days.

19:37 - 19:49

Shasti: So make sure you're grabbing your notes, pen, anything that would help you take notes, and grab your coffee, tea, anything that would make you comfortable for the next couple of hours, and we'll quickly get started.

19:49 - 20:12

Shasti: With that said, we have our amazing mentor in the session already. You met him yesterday, so we have Jitesh back today to teach us on how AI agents can be... make use of... to automate most of her manual work. With that said, Jitesh, welcome back. Folks, let's give a very, very warm welcome to Jitesh in the chat. And Jitesh, with that said, I'm leaving the session to you for the next couple of hours.

20:12 - 20:13

Jitesh Dugar: Alright.

20:13 - 20:16

Jitesh Dugar: Thank you, thank you, Shasti. Hello, everyone.

20:17 - 20:19

Jitesh Dugar: I am so excited to meet everyone.

20:19 - 20:22

Jitesh Dugar: On the second day of learning automations.

20:22 - 20:23

Jitesh Dugar: and building AI agents.

20:24 - 20:25

Jitesh Dugar: No.

20:27 - 20:33

Jitesh Dugar: Before I get started, few rules, right? Stuff that I spoke about yesterday and

20:33 - 20:40

Jitesh Dugar: If you've attended yesterday's session, you already know I'm here to borrow two and a half hours of your undivided attention, full undivided attention.

20:41 - 20:42

Jitesh Dugar: And...

20:42 - 20:55

Jitesh Dugar: In return for that, I promise to you, you'll walk away with some very valuable knowledge that you can implement. And let me tell you, today's knowledge will be possibly one level up compared to what we learned yesterday.

20:55 - 21:14

Jitesh Dugar: Okay? So... and I can... I can tell you for a fact that this is something that you can go to people outside and talk about, you can brag about it, saying, I have built an agent that does this. What it is going to do, we will talk about, okay? So we will go ahead and start exploring those kind of stuff. So...

21:14 - 21:19

Jitesh Dugar: With that, Yes, it will be slightly advanced level.

21:19 - 21:22

Jitesh Dugar: With that, And by the way, I have...

21:22 - 21:30

Jitesh Dugar: I... if you attended yesterday's session, you would know that I have two slightly bad habits. I ask too... too many questions, and I make a ton of mistakes.

21:31 - 21:32

Jitesh Dugar: And I need your help.

21:33 - 21:42

Jitesh Dugar: to answer the questions for me, and help me identify mistakes that I'm making during the session, okay? You need to look carefully. If I'm making mistakes, please stop me right there.

21:43 - 21:44

Jitesh Dugar: No.

21:46 - 21:51

Jitesh Dugar: Alright, perfect. With that... I will go ahead and...

21:51 - 21:58

Jitesh Dugar: Ask you a few questions, and this... this is from the stuff that we discussed yesterday.

21:58 - 22:01

Jitesh Dugar: Now, the first question And you need to tell me.

22:03 - 22:06

Jitesh Dugar: We learned something about AI agents yesterday.

22:07 - 22:10

Jitesh Dugar: Could you tell me what is an AI agent?

22:11 - 22:18

Jitesh Dugar: or rather, let me simplify it further, what are the three essential components of an AI agent that we spoke about yesterday?

22:21 - 22:31

Jitesh Dugar: Yes, I can see the answers are coming. Wow. All the answers are in absolute sync. This is amazing. Yes. The first component of an AI agent is LLM,

22:31 - 22:35

Jitesh Dugar: If people are saying brain, you are also correct, LLM or brain.

22:35 - 22:37

Jitesh Dugar: Or sometimes also a chat model.

22:38 - 22:38

Jitesh Dugar: Okay?

22:39 - 22:42

Jitesh Dugar: All these three things. And the second one.

22:42 - 22:45

Jitesh Dugar: Yes, you guys are right. The second one is memory.

22:48 - 22:54

Jitesh Dugar: And the third one is... tools to execute. Yes, you guys are right.

22:56 - 22:57

Jitesh Dugar: No.

22:58 - 23:02

Jitesh Dugar: Brain gives AI agent an ability to think.

23:02 - 23:10

Jitesh Dugar: to process information, to understand human language, or natural language. It is to remember, memory is to remember.

23:11 - 23:13

Jitesh Dugar: And tools is to execute.

23:14 - 23:19

Jitesh Dugar: There is one more important aspect on AI Agent which I did not talk about yesterday.

23:20 - 23:22

Jitesh Dugar: And that aspect is called...

23:25 - 23:32

Jitesh Dugar: A system prompt. Every AI agent needs a system prompt or system instructions.

23:33 - 23:34

Jitesh Dugar: Let me write it down here.

23:35 - 23:40

Jitesh Dugar: System prompt, or system instructions, system message, all of these are same thing.

23:40 - 23:44

Jitesh Dugar: Do you guys understand what a system prompt or system instructions is?

23:45 - 23:47

Jitesh Dugar: Do you know what a system prompt?

23:48 - 23:53

Jitesh Dugar: Goal, okay, okay? No, okay, that's okay. I will tell you what it is.

23:53 - 23:58

Jitesh Dugar: System prompt or system instructions are like guidelines.

23:58 - 24:05

Jitesh Dugar: They are guidelines that you give to the AI agent to help it understand what it should do.

24:05 - 24:11

Jitesh Dugar: What are the... what is it that... how it should behave? Okay? What is it that it should do? What is it that it should not do?

24:12 - 24:21

Jitesh Dugar: And if you do not understand that also, I have an even easier way to understand this. Do you know, as human... as human resources.

24:21 - 24:24

Jitesh Dugar: We have something called a job description.

24:25 - 24:40

Jitesh Dugar: All of us, if you're working at an organization, you will have a job description. If you are doing something related to marketing, be it sales, human resources, and if you're working at an organization, you will have a job description that'll tell you what are the things that you should do.

24:40 - 24:44

Jitesh Dugar: And system prompt is the job description for an AI agent.

24:46 - 24:48

Jitesh Dugar: Job description for an AI agent.

24:51 - 24:56

Jitesh Dugar: Okay? And... If you want an even better understanding of it.

24:56 - 25:06

Jitesh Dugar: I want to show a simple exercise to all of you. If I go to ChatGPT, everyone, I hope I have your attention on my screen. If you go to ChatGPT, and if you say.

25:07 - 25:11

Jitesh Dugar: how do I hack my neighbor's Wi-Fi?

25:13 - 25:15

Jitesh Dugar: What do you think it'll answer?

25:15 - 25:18

Jitesh Dugar: Will it tell me the steps to hack into a Wi-Fi network?

25:21 - 25:22

Jitesh Dugar: Wow.

25:24 - 25:28

Jitesh Dugar: I think majority of you, 90 of you would say it will not let us do that, but

25:29 - 25:30

Jitesh Dugar: Yes, that's right. Yes.

25:31 - 25:35

Jitesh Dugar: I enter it, And the moment I do that, it will say.

25:36 - 25:45

Jitesh Dugar: I can't help you break into or bypass security of a neighbor's Wi-Fi without their permission. Because this is unethical, it could be illegal also.

25:46 - 25:46

Jitesh Dugar: Okay?

25:47 - 25:58

Jitesh Dugar: And it has been added as a guideline for ChatGPT's agent, saying you are not supposed to help anyone do something illegal or unethical.

25:58 - 26:05

Jitesh Dugar: You can try asking it any unethical stuff, right? How do I rob a bank? How do I do this? It will not answer it.

26:06 - 26:08

Jitesh Dugar: Okay? Whatever it is.

26:09 - 26:19

Jitesh Dugar: Whatever it is, it is inside the system prompt of ChatGPT saying you should... you're not allowed... you're not supposed to let anyone to do any unethical stuff.

26:19 - 26:22

Jitesh Dugar: So, that is about a system prompt.

26:23 - 26:27

Jitesh Dugar: So, system prompt is a guideline that you give to an AI agent.

26:28 - 26:33

Jitesh Dugar: With that, this is a recap from yesterday. Most of it is a recap from yesterday, and...

26:34 - 26:37

Jitesh Dugar: I want to ask another interesting question for you.

26:37 - 26:40

Jitesh Dugar: Which is, imagine you have two people.

26:41 - 26:46

Jitesh Dugar: There's Person 1, Name is possibly Jack, okay?

26:46 - 26:48

Jitesh Dugar: And then there is Person 2.

26:51 - 26:52

Jitesh Dugar: No.

26:52 - 26:56

Jitesh Dugar: For these two people, the second person's name is, maybe, Jill.

26:58 - 26:59

Jitesh Dugar: No.

27:00 - 27:10

Jitesh Dugar: I want you all to imagine a situation where both Person 1 and Person 2, Jack and Jill, they have the same level of intelligence, they have the same level of intelligence.

27:10 - 27:15

Jitesh Dugar: And... The only difference that you have between them is.

27:15 - 27:19

Jitesh Dugar: For Jill, you're giving them a book.

27:19 - 27:25

Jitesh Dugar: You're giving them a book. Give them a book, and you're giving as an instruction, saying.

27:26 - 27:32

Jitesh Dugar: To answer any question, Just look up the book first.

27:33 - 27:40

Jitesh Dugar: And for Person 1, you are saying, Answer it using your brain.

27:41 - 27:44

Jitesh Dugar: And you would also say, never say, I don't know.

27:47 - 27:48

Jitesh Dugar: Okay?

27:49 - 27:50

Jitesh Dugar: I will repeat what we are doing.

27:51 - 28:04

Jitesh Dugar: We are considering two people with the same level of intelligence, and for person two, we are giving them a book, and we are saying, whenever you're supposed to answer any question, just look up the book, and then give me the answer. That is what we are telling them.

28:04 - 28:05

Jitesh Dugar: No.

28:06 - 28:15

Jitesh Dugar: I have a question to all of you. Whenever we ask questions to Person 1, Jack, or Jill, who do you think will excel? Who do you think will give us better answers?

28:20 - 28:24

Jitesh Dugar: Yes, person 2. Jill will get... give us better answers.

28:24 - 28:25

Jitesh Dugar: Okay?

28:26 - 28:29

Jitesh Dugar: Okay, let us start with something, let us start with a few questions.

28:30 - 28:39

Jitesh Dugar: And, possibly the first question can be something like, If my question looks like...
How many grams in

28:40 - 28:42

Jitesh Dugar: In 1KG, okay?

28:42 - 28:46

Jitesh Dugar: Do you think Person 1 and Person 2 both will answer it correctly?

28:47 - 28:48

Jitesh Dugar: I think yes.

28:48 - 28:52

Jitesh Dugar: Right? So both Person 1 and Person 2 will answer it correctly.

28:52 - 28:56

Jitesh Dugar: So, Jack will also give the right answer, Jack... Jill will also give the right answer.

28:57 - 29:01

Jitesh Dugar: Now, let me give you some example of what the book could look like.

29:01 - 29:04

Jitesh Dugar: Let me pull up a book. One minute.

29:19 - 29:27

Jitesh Dugar: Let's say this book is something like this. I pulled up a random book, which is a manual for a microwave oven, okay? So...

29:28 - 29:32

Jitesh Dugar: Let's say this is a book that Jill has access to. Now.

29:33 - 29:41

Jitesh Dugar: If you ask any question to both of them, anything around this book, okay, it could be, how do you set a childproof lock, let's say?

29:41 - 29:46

Jitesh Dugar: Now, if you ask a question saying, how do you set a childproof lock?

29:51 - 29:56

Jitesh Dugar: Who do you think will give the right answer? Will Jack give a right answer, or Jill give a right answer?

29:56 - 30:00

Jitesh Dugar: I think it's very obvious Jill will give us the right answer, because Jill has access to that book.

30:01 - 30:04

Jitesh Dugar: So, most likely, Jack will not give us the right answer.

30:05 - 30:08

Jitesh Dugar: Okay? And Jill will give us a right answer in this case.

30:09 - 30:13

Jitesh Dugar: So, Jill excels, which is something that you're all clear with. Now.

30:14 - 30:22

Jitesh Dugar: I want all of you to understand what is it that Jill did whenever it got a question. Now, we are about to get into something interesting.

30:22 - 30:25

Jitesh Dugar: So, what did Jill do whenever it got a question?

30:27 - 30:29

Jitesh Dugar: Whenever Jill got a question saying.

30:30 - 30:32

Jitesh Dugar: how to set a child lock. So...

30:33 - 30:37

Jitesh Dugar: What Jill did is, Jill has access to a book on one side, okay?

30:37 - 30:39

Jitesh Dugar: And Jill has the question on the other side.

30:39 - 30:48

Jitesh Dugar: how to set a child lock. So what Jill would do is, Jill would go to page number 32 on this book, okay, page number 32 on this book.

30:49 - 30:50

Jitesh Dugar: Where is that?

30:51 - 30:53

Jitesh Dugar: Somewhere around here, okay?

30:53 - 30:59

Jitesh Dugar: Goes to page 32 on the book, there is something around childproof lock, how to set up the lock.

30:59 - 31:05

Jitesh Dugar: And then... That would be the first step. So, the first step that Jill would follow would be...

31:06 - 31:09

Jitesh Dugar: Turn to the relevant page.

31:11 - 31:14

Jitesh Dugar: Turn to the relevant page, or retrieve that page, okay?

31:15 - 31:25

Jitesh Dugar: Once Jill has turned to that relevant page, what would Jill do? Jill will look at the information on that relevant page, look at the question again.

31:26 - 31:38

Jitesh Dugar: Okay? How to set a child log, and try to mix both of them, and then try to get an answer to it. Okay? And what Jill does next is interesting. So what Jill would do next is.

31:38 - 31:40

Jitesh Dugar: Jill would go ahead and...

31:43 - 31:45

Jitesh Dugar: Read the page.

31:46 - 31:48

Jitesh Dugar: Along with the question.

31:49 - 31:52

Jitesh Dugar: That's what Jill would do next. That is step number two.

31:52 - 32:00

Jitesh Dugar: And step number 3, everyone, is... now, Jill has access to that page, and Jill has the question ready in front of

32:01 - 32:02

Jitesh Dugar: Him or her.

32:02 - 32:09

Jitesh Dugar: And now, Jill would now finally look at the question, look at the page, and then give you the answer in plain English.

32:10 - 32:16

Jitesh Dugar: So, it will... Give the answer in plain English.

32:16 - 32:21

Jitesh Dugar: Do you all agree that this is how someone with an access to a book will do?

32:21 - 32:24

Jitesh Dugar: They will turn to the relevant page.

32:24 - 32:30

Jitesh Dugar: They will read the question along with the page, and then they will give the answer.

32:32 - 32:36

Jitesh Dugar: Now, No.

32:37 - 32:41

Jitesh Dugar: So... And I'm about to talk about something very important.

32:41 - 32:47

Jitesh Dugar: In the world of AI Agent, not sure if you've heard of a term, called RAG.

32:49 - 32:54

Jitesh Dugar: There is a term called RAG in the world of AI Agent. Do you know what RAG stands for?

32:56 - 33:00

Jitesh Dugar: It stands for retrieval, augmented.

33:02 - 33:16

Jitesh Dugar: Generation. That's what RAG stands for. It's a very important concept, we will talk about it in much more detail today, but this is what it stands for. And these theory steps that we have done, if you think about it.

33:16 - 33:19

Jitesh Dugar: These three steps are exactly what

33:19 - 33:23

Jitesh Dugar: these do. So, this step, one.

33:23 - 33:27

Jitesh Dugar: It's called retrieval, which means we are retrieving the right page.

33:28 - 33:30

Jitesh Dugar: We are retrieving the right page.

33:30 - 33:37

Jitesh Dugar: Step number two is we are augmenting the information from that page with our question.

33:37 - 33:40

Jitesh Dugar: So, that is augment, augment, augmenting.

33:41 - 33:43

Jitesh Dugar: or augment.

33:44 - 33:49

Jitesh Dugar: And, step number 3, Is generating the answer.

33:50 - 33:52

Jitesh Dugar: So, step number 3 is generate.

33:53 - 34:01

Jitesh Dugar: So this exact series of steps that Jill followed to get the answer is what retrieval augmented generation is.

34:04 - 34:06

Jitesh Dugar: Is everyone able to relate to this?

34:08 - 34:13

Jitesh Dugar: Is it looking clear? Drop 100 on chat if this is looking clear to everyone.

34:15 - 34:16

Jitesh Dugar: Okay?

34:17 - 34:20

Jitesh Dugar: Okay, perfect. Now... With that.

34:22 - 34:30

Jitesh Dugar: Let us do a small exercise again. Let us do a small exercise again. I want all of you to help me answer a question, okay?

34:30 - 34:40

Jitesh Dugar: Imagine... Imagine if I go to ChatGPT, and if I come here, and if I upload that entire of...

34:40 - 34:42

Jitesh Dugar: Okay, let me download it first.

34:42 - 34:46

Jitesh Dugar: Let me download the whole thing, and put it on my...

34:48 - 34:52

Jitesh Dugar: I will go ahead and upload this entire

34:52 - 34:57

Jitesh Dugar: manual, the microwave manual that I had, I will upload the whole thing, and I will ask a question.

34:57 - 35:01

Jitesh Dugar: How to set a... Child Safe Lock.

35:03 - 35:11

Jitesh Dugar: Answer... from the... attached book.

35:11 - 35:18

Jitesh Dugar: Answer from the attached book. This is one example. And the second example, I'll do the same, similar stuff.

35:18 - 35:21

Jitesh Dugar: Where I would... I will say...

35:24 - 35:27

Jitesh Dugar: Yeah, so in this case, I will say.

35:27 - 35:32

Jitesh Dugar: How to... Set a child-safe lock.

35:34 - 35:38

Jitesh Dugar: Answer based on the information below.

35:39 - 35:47

Jitesh Dugar: Okay? And that information will come from this same thing. So I will copy the content of this entire page.

35:52 - 35:56

Jitesh Dugar: Yeah, I'll copy the random information from an entire page, I'll put it here.

35:56 - 35:58

Jitesh Dugar: And I will send both of these.

35:59 - 36:02

Jitesh Dugar: So this is Situation 1, and this is Scenario 2.

36:03 - 36:07

Jitesh Dugar: Okay? I will draw it on a whiteboard, what I have done, okay?

36:07 - 36:09

Jitesh Dugar: Scenario 1.

36:11 - 36:12

Jitesh Dugar: And Scenario 2.

36:13 - 36:17

Jitesh Dugar: In Scenario 1, what we did was, we sent the entire book.

36:20 - 36:25

Jitesh Dugar: And in Scenario 2, Send only specific page.

36:26 - 36:28

Jitesh Dugar: Which one do you think is better?

36:28 - 36:30

Jitesh Dugar: Scenario 1 or Scenario 2?

36:34 - 36:36

Jitesh Dugar: Some of you are saying 1, 2, okay?

36:37 - 36:47

Jitesh Dugar: Only specific. Let us find out what happened, okay? And why Scenario 2 is possibly better, or maybe Scenario 1 is better, I don't know yet, okay? Now.

36:47 - 36:51

Jitesh Dugar: This has given us an answer, and this has also given us an answer.

36:51 - 36:58

Jitesh Dugar: And they all look similar. The actual answer is still similar. Now, If you look at it.

36:59 - 37:08

Jitesh Dugar: Scenario 1 possibly processed about, maybe approximately... Maybe 40,000 words.

37:08 - 37:14

Jitesh Dugar: I don't know, I'm just guessing it. Maybe around 40,000 words is what scenario? One processed.

37:15 - 37:19

Jitesh Dugar: Scenario 2... Processed possibly close to 500 words.

37:21 - 37:26

Jitesh Dugar: And... If you talk about the speed.

37:27 - 37:34

Jitesh Dugar: Scenario 1, because it is processing more information at once, the speed will be slow.

37:35 - 37:36

Jitesh Dugar: Slow speed.

37:36 - 37:38

Jitesh Dugar: And Scenario 2?

37:38 - 37:40

Jitesh Dugar: It is faster.

37:42 - 37:45

Jitesh Dugar: Because it is processing less information.

37:45 - 37:46

Jitesh Dugar: And?

37:47 - 37:50

Jitesh Dugar: And... The cost between these?

37:50 - 37:51

Jitesh Dugar: Scenario 1?

37:52 - 37:53

Jitesh Dugar: It is costly.

37:54 - 37:59

Jitesh Dugar: And when you say costly, it is probably about 80x costly, compared to Scenario 2.

38:00 - 38:02

Jitesh Dugar: So this is cheaper.

38:02 - 38:04

Jitesh Dugar: Compared to Scenario 1.

38:05 - 38:13

Jitesh Dugar: And this is on the token consumption. Now, if you're wondering, how come... how come this is cheaper compared to this, that's because...

38:14 - 38:15

Jitesh Dugar: Because...

38:15 - 38:29

Jitesh Dugar: or the way these LLMs charge, right now, in case of ChatGPT, we pay a fixed subscription fee, the cost is, I mean, it is all borne by the subscription itself, but the way they charge is based on the input tokens.

38:30 - 38:32

Jitesh Dugar: So, whatever we send in the input.

38:33 - 38:38

Jitesh Dugar: Whatever we send in the input, that consumes input token cost.

38:39 - 38:49

Jitesh Dugar: So, in this case, we sent 500 words, possibly. In this example, we possibly sent 40,000 words. So, in case of the... talking about the cost aspect, this would consume

38:50 - 38:51

Jitesh Dugar: 80 times the cost.

38:52 - 39:01

Jitesh Dugar: That's what happens. And that is the reason it is always better to give your AI agent context for things that it cares about.

39:03 - 39:14

Jitesh Dugar: Give the agent context of things that it cares about. So, in this case, Scenario 2 is more superior compared to Scenario 1, because it has relevant context.

39:14 - 39:19

Jitesh Dugar: On top of that, it is more likely to give you more accurate information also.

39:20 - 39:22

Jitesh Dugar: And it can work faster.

39:24 - 39:31

Jitesh Dugar: Now, and we will see all of these in practice, it's not just me talking right now, we'll see all of these in practice today, okay? Now.

39:34 - 39:38

Jitesh Dugar: With that information, With that information.

39:38 - 39:44

Jitesh Dugar: We want to solve a business problem statement today. We want to solve a business problem today.

39:45 - 39:49

Jitesh Dugar: And... As problem solvers, You need to help me?

39:49 - 39:54

Jitesh Dugar: go through or solve this problem. So, our problem statement goes like this.

39:57 - 40:03

Jitesh Dugar: A problem statement is... Imagine you are running a company.

40:04 - 40:08

Jitesh Dugar: And the name of that company is an online bakery, let's say.

40:09 - 40:13

Jitesh Dugar: online bakery business. Let's say you're running an online bakery business.

40:16 - 40:19

Jitesh Dugar: And the name of the brand is, let's say, Bakehouse.

40:21 - 40:22

Jitesh Dugar: And?

40:23 - 40:27

Jitesh Dugar: You get a lot of customer queries via email.

40:27 - 40:31

Jitesh Dugar: gets... get customer... Queries.

40:32 - 40:33

Jitesh Dugar: Viaia.

40:34 - 40:40

Jitesh Dugar: And you have a customer support team, That answers those queries.

40:41 - 40:44

Jitesh Dugar: The customer support team will answer all of those queries for you.

40:45 - 40:46

Jitesh Dugar: No.

40:47 - 40:50

Jitesh Dugar: One fine day, you had...

40:51 - 40:59

Jitesh Dugar: Possibly a very good day where your sales, or maybe a very good week, where your sales suddenly went up, which is really good news.

41:00 - 41:09

Jitesh Dugar: But... With that, possibly, Possibly what also happened was, you suddenly got 10,000 customer support emails.

41:10 - 41:15

Jitesh Dugar: Which means 10,000 customers sent you an email on one single day, and you're supposed to

41:15 - 41:18

Jitesh Dugar: Write an answer to all of those emails.

41:21 - 41:26

Jitesh Dugar: And... If you have a situation like this, what would you do?

41:28 - 41:34

Jitesh Dugar: Imagine how much time would it take for you to answer one single email from a customer?

41:35 - 41:36

Jitesh Dugar: A rough estimate, everyone?

41:38 - 41:42

Jitesh Dugar: Someone's saying 15 minutes. No, 15 minutes, I think, is too... a lot.

41:42 - 41:50

Jitesh Dugar: 5 minutes is fair, maybe if you're too efficient, maybe 3 to 5 minutes. Let's assume 5 minutes, okay? 5 minutes per email.

41:50 - 41:54

Jitesh Dugar: To answer 10,000 customer emails, how much time would it take?

41:57 - 42:02

Jitesh Dugar: 50,000 minutes, and how much is 50,000 minutes in hours?

42:06 - 42:08

Jitesh Dugar: Okay, we need some really good math brains right now.

42:12 - 42:15

Jitesh Dugar: How much is that? 833 hours, yes.

42:16 - 42:22

Jitesh Dugar: Wonderful, okay. 833 hours to reply to those 10,000 customer support emails.

42:22 - 42:25

Jitesh Dugar: What do you think is a better way to solve it?

42:26 - 42:29

Jitesh Dugar: As problem solvers, how can we do better?

42:29 - 42:34

Jitesh Dugar: If... Manually, humans have to reply. It would take them 34 days.

42:34 - 42:36

Jitesh Dugar: To reply to these emails.

42:36 - 42:42

Jitesh Dugar: For one person. Of course, if there are more people, it would take less, right? You can either hire more people.

42:43 - 42:45

Jitesh Dugar: Or, you can automate.

42:46 - 42:48

Jitesh Dugar: you can automate, okay? So...

42:49 - 43:01

Jitesh Dugar: Let us think about how do we go about automating, and we'll think whether we need an AI agent or not. We'll think about it. For now, we're considering a way where we can automate replies to the emails.

43:02 - 43:03

Jitesh Dugar: No.

43:05 - 43:08

Jitesh Dugar: Do you know what the first step of an automation is?

43:09 - 43:12

Jitesh Dugar: What should we do when automating it?

43:13 - 43:17

Jitesh Dugar: We should first go ahead and plan it out. Draw it on a whiteboard.

43:17 - 43:22

Jitesh Dugar: Okay, we would go ahead and draw it on a whiteboard, what we are supposed to do.

43:22 - 43:23

Jitesh Dugar: So...

43:26 - 43:33

Jitesh Dugar: So, on the whiteboard, we'll have to draw a flow chart of what we are supposed to do. So, on that flowchart.

43:34 - 43:35

Jitesh Dugar: What should we do?

43:35 - 43:44

Jitesh Dugar: So, the first step If you're supposed to build an automatic email replying automation, what should be the trigger?

43:47 - 43:52

Jitesh Dugar: The trigger, which will be the first step, Yes.

43:53 - 43:59

Jitesh Dugar: So, trigger will be... An incoming email. Yes, you guys are correct. Perfect.

44:00 - 44:06

Jitesh Dugar: Trigger will be an incoming email, where anytime you get an incoming email, that should trigger your automation.

44:07 - 44:08

Jitesh Dugar: No.

44:09 - 44:12

Jitesh Dugar: When you get an incoming email, what should happen next?

44:14 - 44:16

Jitesh Dugar: What should we do?

44:18 - 44:20

Jitesh Dugar: Yes, Akshay, that is correct.

44:24 - 44:26

Jitesh Dugar: Yes, I can see so many ideas here.

44:28 - 44:40

Jitesh Dugar: Acknowledge, yes, acknowledge is also good, but if you're saying acknowledge, which is great, which means you might want to acknowledge to the user, saying, hey, thank you for sending us an email, we'll reply to you within 24 hours.

44:40 - 44:44

Jitesh Dugar: But if automation is handling that, do you agree that

44:45 - 44:50

Jitesh Dugar: The automation will reply to it almost instantly, so there's no point of sending them an acknowledgement.

44:50 - 45:00

Jitesh Dugar: So, let us go ahead and do a classification. So, we will classify We will classify.

45:00 - 45:07

Jitesh Dugar: And the classify step, what it should do is, it should help us identify if it is related to customer support or not.

45:08 - 45:10

Jitesh Dugar: Customer support or not?

45:11 - 45:14

Jitesh Dugar: That's what we will... classify with.

45:14 - 45:20

Jitesh Dugar: For example, it could be a spam email, it could be a marketing email that we are not supposed to reply to.

45:20 - 45:26

Jitesh Dugar: So, we will add a classification step to classify whether it is spam or not.

45:29 - 45:33

Jitesh Dugar: Next, once you have classified whether it is spam or not.

45:35 - 45:38

Jitesh Dugar: Should we go ahead and create some branches here?

45:39 - 45:40

Jitesh Dugar: Where?

45:41 - 45:45

Jitesh Dugar: The next step will help us check if it is customer support or not.

45:47 - 45:50

Jitesh Dugar: Customer support or not customer support?

45:51 - 45:54

Jitesh Dugar: And based on that, we should decide what to do.

45:54 - 45:58

Jitesh Dugar: If it is related to customer support, we should do something.

45:59 - 46:02

Jitesh Dugar: Remember, CS stands for Customer Support?

46:02 - 46:06

Jitesh Dugar: If it is not related to customer support, we must do something else.

46:08 - 46:15

Jitesh Dugar: So, if it is customer support, it should go to the branch on top. If it is not customer support, it should go to the branch at bottom, okay?

46:16 - 46:17

Jitesh Dugar: And...

46:17 - 46:26

Jitesh Dugar: If it is not related to customer support, we can possibly say, do nothing, we probably don't want to do anything, we'll just ignore, or ignore, okay?

46:26 - 46:30

Jitesh Dugar: If it is related to customer support, then we have to do a few things.

46:30 - 46:33

Jitesh Dugar: What are those few things? We'll have to decide.

46:36 - 46:46

Jitesh Dugar: And those few things will be... Those few things will be... Understand the customer's message.

46:47 - 46:48

Jitesh Dugar: That's the first thing.

46:49 - 46:54

Jitesh Dugar: The second one will be... Look up our knowledge base.

46:56 - 46:59

Jitesh Dugar: To understand what is the policy that

46:59 - 47:09

Jitesh Dugar: Bakehouse follows, what are the refund policies? What are the delivery policies? How does someone place an order, what are the store opening hours, and all of those things.

47:09 - 47:13

Jitesh Dugar: So, whenever someone's asking us a question, we should

47:13 - 47:17

Jitesh Dugar: Go to the knowledge base, and get an answer to that question.

47:17 - 47:20

Jitesh Dugar: And then, it should prepare a reply.

47:21 - 47:28

Jitesh Dugar: Because now we know that we have a question, and we have read the knowledge base, and we are supposed to prepare a reply to it.

47:31 - 47:34

Jitesh Dugar: And, the last step, Is?

47:35 - 47:37

Jitesh Dugar: Send the reply.

47:38 - 47:39

Jitesh Dugar: to the customer.

47:44 - 47:52

Jitesh Dugar: Is it making sense, these four steps, everyone? On a scale of 1 to 10, is it making sense on all of the steps that we've spoken about so far?

47:52 - 47:54

Jitesh Dugar: 10, yes, okay, perfect.

47:55 - 47:56

Jitesh Dugar: Perfect, thank you.

47:57 - 47:58

Jitesh Dugar: Great.

47:58 - 48:03

Jitesh Dugar: Yes, please keep that energy and the attention level high. We are about to get into very interesting stuff.

48:05 - 48:05

Jitesh Dugar: No.

48:06 - 48:11

Jitesh Dugar: And what do you think is that block here that'll do all of these for us?

48:11 - 48:18

Jitesh Dugar: It should understand the customer's message, it should look up our knowledge base, and it should prepare a reply, and it should send a reply.

48:19 - 48:22

Jitesh Dugar: Will it be an LLM or an AI agent?

48:23 - 48:25

Jitesh Dugar: LLM or AI agent?

48:27 - 48:33

Jitesh Dugar: Okay, I can see about 40% people saying LLM, remaining 60% saying AI agent.

48:34 - 48:36

Jitesh Dugar: Okay, people who are saying LLM,

48:39 - 48:43

Jitesh Dugar: No, it's not the right answer. The right answer is AI Agent, because

48:43 - 48:49

Jitesh Dugar: LLMs can predict the word for you, which means LLMs can do this part.

48:49 - 48:54

Jitesh Dugar: LMs, sorry, LMs can do this part. But if you want to send a reply, you really need

48:54 - 48:57

Jitesh Dugar: An ability to execute something.

48:57 - 48:59

Jitesh Dugar: And that's why you need an AI agent.

49:00 - 49:05

Jitesh Dugar: So this thing will be an AI agent that will reply to the customer.

49:05 - 49:07

Jitesh Dugar: By looking up the knowledge base.

49:09 - 49:15

Jitesh Dugar: LLM is just the brain. AI Agent will include a brain, a memory, and it can use tools also.

49:16 - 49:18

Jitesh Dugar: So that's why we need NEI agent.

49:19 - 49:25

Jitesh Dugar: I will just repeat the workflow. Our workflow will start with an incoming email.

49:25 - 49:31

Jitesh Dugar: Which means whenever we get an incoming email on our inbox, we would start our automation.

49:31 - 49:41

Jitesh Dugar: We would have a classification step that will read the incoming email and tell us whether the incoming email is related to customer support or not related to customer support.

49:42 - 49:45

Jitesh Dugar: And based on that, we create a branch.

49:46 - 49:48

Jitesh Dugar: Based on that, we create a branch.

49:48 - 49:51

Jitesh Dugar: And that branch will split into two parts.

49:51 - 49:54

Jitesh Dugar: If it is customer support, it'll go into this part.

49:54 - 49:57

Jitesh Dugar: If it is not customer support, we'll go into this part.

49:58 - 50:08

Jitesh Dugar: And if it is not customer support, it will do nothing, it will ignore. If it is customer support, it will send it to an AI agent that will do all of these things for us.

50:10 - 50:12

Jitesh Dugar: Net will do all of these things for us.

50:13 - 50:14

Jitesh Dugar: No.

50:15 - 50:32

Jitesh Dugar: Now, one very good question. Why not go ahead and create a branch from step to itself? That's... that's how it works also, but the thing is, you need an AI to first classify, and then create a branch. You really cannot create a branch directly from the LLM.

50:33 - 50:34

Jitesh Dugar: From the classify step.

50:35 - 50:40

Jitesh Dugar: And question for all of you. This is based on the knowledge from yesterday, and maybe I broke it also.

50:41 - 50:45

Jitesh Dugar: What steps should this be? Should this be an AI agent or an LLM?

50:51 - 50:58

Jitesh Dugar: Yes, this will be an LLM. LLMs are very well suited for classify-related actions.

50:58 - 51:02

Jitesh Dugar: When you do anything related to classify, LLMs are great.

51:04 - 51:07

Jitesh Dugar: So, this is the workflow that we will be building today.

51:08 - 51:09

Jitesh Dugar: No.

51:10 - 51:11

Jitesh Dugar: Now...

51:11 - 51:21

Jitesh Dugar: Let me pull up the... an example of a knowledge base that I have for today. How it will read the knowledge base, that will be a very interesting aspect.

51:21 - 51:24

Jitesh Dugar: And let me tell you how important this session is.

51:24 - 51:32

Jitesh Dugar: Stuff like RAC customer support agents have found the highest level of adoption in any industry.

51:32 - 51:38

Jitesh Dugar: It is one of the most adopted AI solutions that so many industries have adopted today.

51:38 - 51:46

Jitesh Dugar: Because of the kind of accuracy that it provides. And you have seen, you have all seen, that same example. Jack and Jill example.

51:47 - 51:48

Jitesh Dugar: Where is that? Yeah, there it is.

51:49 - 51:49

Jitesh Dugar: So...

51:50 - 51:57

Jitesh Dugar: A huge problem in the world of AI is, AIs will never say, I don't know. They will confidently give you a wrong answer.

51:57 - 52:02

Jitesh Dugar: And giving them access to a book, or rather, a knowledge base.

52:03 - 52:13

Jitesh Dugar: Rather, a knowledge base makes the AI to give you accurate and very correct answers, and that is where a RAC system is extremely useful and important.

52:16 - 52:23

Jitesh Dugar: So with this exercise that I wanted to show you was for... to help you emphasize or understand how important RAC systems are.

52:24 - 52:30

Jitesh Dugar: AI will confidently give you incorrect answers if it is not supported with some kind of a

52:30 - 52:32

Jitesh Dugar: Some kind of a book.

52:32 - 52:46

Jitesh Dugar: Okay? And this is the retrieval augmented generation process, where we looked up the relevant page, read the page along with the question, give them the answer in plain English. We will dive much deeper into the concept of RAG.

52:47 - 52:54

Jitesh Dugar: Because, technically, it's not just this much, there is a ton of other stuff that happens behind the scenes, and we will learn all of that today.

52:57 - 52:57

Jitesh Dugar: No.

52:58 - 53:04

Jitesh Dugar: With that, let me pull up... A sample knowledge base.

53:07 - 53:11

Jitesh Dugar: There you go. Yeah, I have a sample knowledge base for Bakehouse customer support.

53:11 - 53:12

Jitesh Dugar: policy.

53:13 - 53:15

Jitesh Dugar: And this is that book, or handbook.

53:16 - 53:27

Jitesh Dugar: that I will give to our agent, and I will tell the agent there are close to 9 pages right now, but remember, this could have been 10,000 pages also, and it could work with the same level of

53:27 - 53:31

Jitesh Dugar: accuracy with even 10,000 pages, that's the beauty of RAG.

53:32 - 53:35

Jitesh Dugar: So, a lot of stuff about the company, like...

53:36 - 53:48

Jitesh Dugar: What the company does? It sells fresh breads, cakes, cookies, savory snacks, store timings, online order timings, delivery areas, delivery charges, contact channels, product categories.

53:49 - 53:54

Jitesh Dugar: How do you place an order? What are the lead times? How do you modify an order? What are the delivery slots?

53:55 - 53:58

Jitesh Dugar: Refund policy, all of these things are mentioned.

53:58 - 54:09

Jitesh Dugar: So, if anyone has a question, any customer asks a question, hey, I placed an order, my delivery was delayed by 2 hours, what should I do? It will go to this part.

54:09 - 54:18

Jitesh Dugar: And it'll say, if the delivery was more than 2 hours late and the customer accepted the order, the refund of the delivery charge, plus 10% Goodwill credit will be given.

54:19 - 54:23

Jitesh Dugar: So, the AI agent will be able to give the exact, accurate answer.

54:24 - 54:26

Jitesh Dugar: By looking up this handbook.

54:27 - 54:30

Jitesh Dugar: And this is what RAG will help us do.

54:31 - 54:32

Jitesh Dugar: No.

54:32 - 54:35

Jitesh Dugar: How does it work? We will now talk about.

54:35 - 54:38

Jitesh Dugar: But before that, let's start building some part of it.

54:38 - 54:40

Jitesh Dugar: on anything.

54:46 - 54:49

Jitesh Dugar: So, on NA10, you have all seen this interface.

54:50 - 54:54

Jitesh Dugar: To build a workflow, we click on this orange button that says Create Workflow.

54:56 - 55:00

Jitesh Dugar: It will open up a blank new workflow canvas for you.

55:01 - 55:05

Jitesh Dugar: And let us give it a name, we'll call it... Bakehouse?

55:05 - 55:08

Jitesh Dugar: Customer Support Agent.

55:09 - 55:11

Jitesh Dugar: That's the name for our workflow.

55:11 - 55:12

Jitesh Dugar: No.

55:13 - 55:17

Jitesh Dugar: The first step, everyone, is... Click on the plus button.

55:18 - 55:24

Jitesh Dugar: And it will give you an option to choose what will trigger your workflow.

55:24 - 55:27

Jitesh Dugar: Can you tell me what should I select here?

55:27 - 55:31

Jitesh Dugar: And I 100% know that everyone will give me a right answer.

55:32 - 55:34

Jitesh Dugar: Yes, email, that's correct.

55:35 - 55:44

Jitesh Dugar: Yes, email. It depends on what email you're using. I use Gmail, so I will choose Gmail as my email trigger step. So I will choose Gmail.

55:45 - 55:50

Jitesh Dugar: If you're choosing Outlook or anything like that, you are free to choose those also. Those are all supported.

55:50 - 55:53

Jitesh Dugar: Let's select Gmail right now, and there is a trigger that says.

55:53 - 56:00

Jitesh Dugar: OnMessage Received, which means whenever you receive a new email, it will trigger the workflow.

56:00 - 56:05

Jitesh Dugar: Now, if you're wondering, Can I do the same thing with, maybe, WhatsApp?

56:05 - 56:08

Jitesh Dugar: Or any other channel, yes, you can do that.

56:09 - 56:13

Jitesh Dugar: You can connect with WhatsApp also. There is a process to do that.

56:13 - 56:16

Jitesh Dugar: But for now, we'll talk about email itself.

56:17 - 56:22

Jitesh Dugar: So you select On Message Received, And, after that.

56:22 - 56:28

Jitesh Dugar: It is now asking... and it is now asking you, give me the credential for your Gmail account.

56:28 - 56:31

Jitesh Dugar: Give me the credential for your Gmail account.

56:31 - 56:37

Jitesh Dugar: and the Gmail account to which I should look up all the incoming emails.

56:37 - 56:40

Jitesh Dugar: Assume this is the Gmail account that we have.

56:40 - 56:44

Jitesh Dugar: And it should read the incoming emails from this Gmail account.

56:44 - 56:45

Jitesh Dugar: So?

56:45 - 56:56

Jitesh Dugar: If you want to authenticate it, you can click on Create New Credential, click on Sign In with Google, there'll be a couple of screens through which you can pass and complete the authentication process.

56:58 - 57:01

Jitesh Dugar: And... Next thing is, it is now asking us.

57:02 - 57:09

Jitesh Dugar: Poll time, which means how frequently would you like the workflow to look for new emails on the inbox?

57:09 - 57:16

Jitesh Dugar: Right now, it's set to every minute, but you can decide to set it to every hour, every day, all of these options. For now, every minute is good enough for us.

57:16 - 57:18

Jitesh Dugar: Then the next option is...

57:18 - 57:26

Jitesh Dugar: Event. Event is message received. Simplify, you can keep it turned on, but let me tell you what Simplify does.

57:26 - 57:30

Jitesh Dugar: Simplify helps you identify if you want to get

57:30 - 57:40

Jitesh Dugar: extremely detailed information about your email, then you can turn off Simplify. But I recommend you to turn on Simplify for now, so that you get simplified version of your output.

57:42 - 57:43

Jitesh Dugar: Okay?

57:43 - 57:44

Jitesh Dugar: No.

57:45 - 57:54

Jitesh Dugar: And how many emails would you like the workflow to process in one execution? Maybe we are setting it to 10, you can decide to increase it to 100 also. That's up to you.

57:56 - 57:59

Jitesh Dugar: There is no cost aspect when you determine the frequency.

57:59 - 58:03

Jitesh Dugar: There is no additional cost for it, because it only processes it

58:04 - 58:06

Jitesh Dugar: When there is a new email.

58:09 - 58:10

Jitesh Dugar: With that...

58:11 - 58:22

Jitesh Dugar: After I have set the trigger, I will click on the button here that says Fetch Test Event, where I would tell the system, go ahead and look it to my inbox and look for a new email.

58:23 - 58:28

Jitesh Dugar: And you know what happened? It said, we did not find data in Gmail to simulate an event.

58:28 - 58:34

Jitesh Dugar: Which means... it did not find any new email on my Gmail inbox.

58:35 - 58:43

Jitesh Dugar: And for that reason, we have to go ahead and send one sample email to ourselves, so that we can simulate a scenario for us.

58:44 - 58:45

Jitesh Dugar: Let me go ahead and do that.

58:56 - 59:01

Jitesh Dugar: Yeah, so I will pull up open my inbox, and I will just write an email.

59:01 - 59:04

Jitesh Dugar: to this Bakehouse customer support.

59:04 - 59:08

Jitesh Dugar: And I would say... I would possibly ask a question

59:10 - 59:13

Jitesh Dugar: My delivery was delayed by more than 2 hours.

59:18 - 59:26

Jitesh Dugar: Hi, Bakehouse team. My recent delivery was delayed by more than 2 hours. This was a very bad experience for me, I just wanted to share a feedback.

59:28 - 59:33

Jitesh Dugar: So, this is a sample email that we are sending, and we would go ahead and send them email.

59:34 - 59:35

Jitesh Dugar: And?

59:36 - 59:39

Jitesh Dugar: After we've sent the email, we would click on Fetch Test Event to see if it has

59:40 - 59:46

Jitesh Dugar: Fetch that email, it still hasn't. We'll do it one more time, it may take a few seconds, or maybe a minute.

59:54 - [01:00:13]

Jitesh Dugar: Yeah, there you go. We have the fetch test event. Upon clicking Fetch Test Event, you have the output. In the output, you can see it says, Hi Bakehouse team, my recent delivery was delayed by more than 2 hours, this was a very bad experience for me. I just wanted to share feedback, and the subject, the signature shows up. And on the output.

[01:00:14] - [01:00:32]

Jitesh Dugar: You have a table view, you have a schema view, and you have a JSON view. These are three views for you to check... look at the same piece of information. So I will switch back to schema view, because I prefer working with the schema view. I just think it's easier to work with, but it's your personal preference in this case.

[01:00:33] - [01:00:37]

Jitesh Dugar: And again, What should we do here?

[01:00:37] - [01:00:41]

Jitesh Dugar: Yesterday, we did something after this step. What was that step?

[01:00:42] - [01:00:44]

Jitesh Dugar: Yes, you guys are right.

[01:00:45] - [01:00:46]

Jitesh Dugar: We should pin the data.

[01:00:47] - [01:00:50]

Jitesh Dugar: You should pin the data, and there is a small pin icon here.

[01:00:51] - [01:00:54]

Jitesh Dugar: That allows you to pin the data. We are pinning the data because...

[01:00:55] - [01:00:58]

Jitesh Dugar: Because, if we refresh this workflow.

[01:00:58] - [01:01:02]

Jitesh Dugar: If I refresh this workflow, we do not want this data to get lost.

[01:01:03] - [01:01:07]

Jitesh Dugar: And that's why we pinned the data, and we can now close it.

[01:01:07] - [01:01:10]

Jitesh Dugar: We have the first step executed.

[01:01:10] - [01:01:14]

Jitesh Dugar: We have the first step executed. With that, this step is completed.

[01:01:17] - [01:01:28]

Jitesh Dugar: You can pin the other steps also, not just the triggers. You can pin the other steps also. Pin allows you to save the execution data, so that even if you refresh this page.

[01:01:28] - [01:01:31]

Jitesh Dugar: Your data still stays safe, and saved.

[01:01:32] - [01:01:41]

Jitesh Dugar: And you can use it in further parts of your workflow. And it gets very useful to... to get some data so that you can build your workflow. That's why pinning data is very useful.

[01:01:41] - [01:01:48]

Jitesh Dugar: Remember, PIN data only works for test executions. When you get an actual email, it will not use the PIN data.

[01:01:50] - [01:01:56]

Jitesh Dugar: With that... We have completed the first trigger step on our workflow.

[01:01:56] - [01:02:01]

Jitesh Dugar: And the next step for us is adding a classify step.

[01:02:01] - [01:02:02]

Jitesh Dugar: Which is LLM.

[01:02:03] - [01:02:05]

Jitesh Dugar: Click on the plus button.

[01:02:05] - [01:02:11]

Jitesh Dugar: And... Click on AI. The first option that it shows, AI, click on it.

[01:02:12] - [01:02:29]

Jitesh Dugar: And there are so many LLMs that you can choose from. You can choose Anthropic, you can choose Google Gemini, any of these, but we use OpenAI a lot of times, so I will be using OpenAI itself. Others are also all okay, you can use others also, that's all up to you.

[01:02:29] - [01:02:30]

Jitesh Dugar: So we'll use OpenAI.

[01:02:31] - [01:02:41]

Jitesh Dugar: And the action that we would select, because we're supposed to send a message to a large language model, the text action here would be message a model.

[01:02:42] - [01:02:44]

Jitesh Dugar: Let's select message or model.

[01:02:45] - [01:02:50]

Jitesh Dugar: And make sure your OpenAI credentials are connected.

[01:02:50] - [01:02:56]

Jitesh Dugar: That means you're giving Ann an ability to use your OpenAI account.

[01:02:58] - [01:03:02]

Jitesh Dugar: Okay? And by the way, I got a very interesting question, and Vibhanshu, you're right.

[01:03:02] - [01:03:10]

Jitesh Dugar: Vibhanshu got to ask me a question, which is, instead of OpenAI, why are you not using the classified Tex DeF? There is a...

[01:03:10] - [01:03:13]

Jitesh Dugar: Classify... text classifier also.

[01:03:13] - [01:03:17]

Jitesh Dugar: You are correct, we can use this also. Okay? But...

[01:03:17] - [01:03:31]

Jitesh Dugar: Text classifier is meant only for classification. We are intentionally using something more generic, so that you can do a lot of other stuff also. This is more generic, and it can be applied to a wider use case.

[01:03:31] - [01:03:37]

Jitesh Dugar: That's why we are using this itself. But I can tell you, Classify will also work very well.

[01:03:40] - [01:03:45]

Jitesh Dugar: After that, I have a recommendation or suggestion to all of you. Do not try to do it alongside.

[01:03:46] - [01:03:50]

Jitesh Dugar: We are going to be talking about something important, and I will be making mistakes.

[01:03:50] - [01:03:53]

Jitesh Dugar: If you're doing it alongside, you'll not help me catch those mistakes.

[01:03:56] - [01:04:00]

Jitesh Dugar: Next, Once you've connected your credentials.

[01:04:00] - [01:04:02]

Jitesh Dugar: You have to choose the model.

[01:04:02] - [01:04:06]

Jitesh Dugar: Do you know what model should we choose here for the classification part?

[01:04:09] - [01:04:17]

Jitesh Dugar: Okay, I can see some answers. Wow, literally everyone's saying mini, okay? This is a concept of a small brain versus a big brain.

[01:04:18] - [01:04:27]

Jitesh Dugar: Okay? If your task is very basic, do not use a pro-brain. Use a mini brain itself, or maybe a nanobrain also.

[01:04:28] - [01:04:34]

Jitesh Dugar: Let's select a 4.1 mini, you can choose a 4.0 Mini also, you can choose 5.4 mini, any of those are okay.

[01:04:35] - [01:04:36]

Jitesh Dugar: Let's choose this.

[01:04:37] - [01:04:40]

Jitesh Dugar: And now... We have an option of adding a prompt.

[01:04:41] - [01:04:43]

Jitesh Dugar: We have an option of adding a prompt.

[01:04:44] - [01:04:47]

Jitesh Dugar: In the prompt, In the prompt.

[01:04:47] - [01:04:51]

Jitesh Dugar: The option here says Fixed an expression.

[01:04:52] - [01:04:55]

Jitesh Dugar: Fixed means I can only provide static values.

[01:04:56] - [01:04:58]

Jitesh Dugar: I can only provide static values.

[01:04:59] - [01:05:03]

Jitesh Dugar: But expression allows you to add dynamic values.

[01:05:03] - [01:05:04]

Jitesh Dugar: Now...

[01:05:04] - [01:05:11]

Jitesh Dugar: Let me start by adding a prompt here, and there's a small little arrow right here that allows us to make this screen a little bigger.

[01:05:12] - [01:05:13]

Jitesh Dugar: So I will click on this arrow.

[01:05:14] - [01:05:17]

Jitesh Dugar: And the screen becomes bigger so that we can start typing here.

[01:05:17] - [01:05:20]

Jitesh Dugar: So, over here is where we'll put the prompt.

[01:05:20] - [01:05:22]

Jitesh Dugar: And I'm going to dictate the prompt.

[01:05:26] - [01:05:40]

Jitesh Dugar: You are a email intent classifier for Bakehouse Team, which is an online bakery business. You're supposed to look at an incoming email and help me classify whether it is related to customer support or not.

[01:05:41] - [01:05:50]

Jitesh Dugar: If it is related to customer support, then reply with CS. If it is not related to customer support, reply with not CS.

[01:05:50] - [01:05:54]

Jitesh Dugar: Do not reply with anything other than CS or not CS.

[01:05:56] - [01:05:58]

Jitesh Dugar: A very basic prompt that we've given.

[01:06:05] - [01:06:07]

Jitesh Dugar: And you can also say.

[01:06:08] - [01:06:25]

Jitesh Dugar: Any queries related to ordering online, getting refund, delivery tracking, store timings, and all of the related areas are related to customer support. Any other queries that are related to marketing emails, spam category emails, those are not customer support emails.

[01:06:26] - [01:06:32]

Jitesh Dugar: So these are the kind of prompts that you can write, and you can, of course, go to a ChatGPT or a Claude and ask it to improve the prompt.

[01:06:35] - [01:06:39]

Jitesh Dugar: Okay? Now... With that, our prompt has been added.

[01:06:39] - [01:06:41]

Jitesh Dugar: And we will close it.

[01:06:41] - [01:06:43]

Jitesh Dugar: And we will go ahead and execute this step.

[01:06:47] - [01:06:49]

Jitesh Dugar: And we are getting a message.

[01:06:49] - [01:06:54]

Jitesh Dugar: And the message says, understood, please provide the incoming email for the classification.

[01:06:55] - [01:06:57]

Jitesh Dugar: Okay, what is this? What should we do?

[01:06:57] - [01:06:59]

Jitesh Dugar: Did I make some mistake?

[01:07:07] - [01:07:10]

Jitesh Dugar: Yes, the dynamic input is missing.

[01:07:10] - [01:07:12]

Jitesh Dugar: Yes, you're right.

[01:07:13] - [01:07:18]

Jitesh Dugar: In this, we have given it a very static prompt, where we're just saying you need to classify as this.

[01:07:19] - [01:07:22]

Jitesh Dugar: And you need to reply with either a CS or not CS.

[01:07:25] - [01:07:27]

Jitesh Dugar: Let me structure it a little better.

[01:07:30] - [01:07:32]

Jitesh Dugar: So we need to now give it the email content.

[01:07:33] - [01:07:38]

Jitesh Dugar: Sorry, this one will be not CS, so this will go like this.

[01:07:42] - [01:07:42]

Jitesh Dugar: Yeah.

[01:07:43] - [01:07:52]

Jitesh Dugar: We are now supposed to give it the email content that it needs to classify. And for that, we should switch to expression so that we are able to

[01:07:52] - [01:07:57]

Jitesh Dugar: make it a dynamic field. So, switch to expression.

[01:07:57] - [01:08:00]

Jitesh Dugar: And then click on this arrow right here, this arrow.

[01:08:00] - [01:08:08]

Jitesh Dugar: it makes the screen bigger, and now you're ready to give it the email content that it should classify. So, you would just say.

[01:08:10] - [01:08:11]

Jitesh Dugar: Email to classify.

[01:08:12] - [01:08:15]

Jitesh Dugar: And that email goes like this. You can give it the subject.

[01:08:15] - [01:08:21]

Jitesh Dugar: And the subject of the email will come from the left-hand side, which will be from the Gmail trigger step, okay?

[01:08:21] - [01:08:26]

Jitesh Dugar: And we will scroll down, look for the subject. There you go, the subject is right here.

[01:08:26] - [01:08:31]

Jitesh Dugar: Okay, and we will go ahead and track this subject and put it right here.

[01:08:32] - [01:08:38]

Jitesh Dugar: And we will then... have to provide the email snippet. Snippet of the email.

[01:08:39] - [01:08:42]

Jitesh Dugar: And the snippet of the email will come from the left-hand side.

[01:08:42] - [01:08:44]

Jitesh Dugar: Drag it and put it right here.

[01:08:44] - [01:08:51]

Jitesh Dugar: And you can view the result on the right-hand side, where you can see what was the... what is the prompt that will now go to

[01:08:51] - [01:08:53]

Jitesh Dugar: the OpenAI step.

[01:08:54] - [01:09:04]

Jitesh Dugar: Now, question, why did I shorten customer support to CS and not CS? That's a personal choice. You can decide to replace this with a yes-no also. You can decide to use

[01:09:05] - [01:09:07]

Jitesh Dugar: The full words also, that's all up to you.

[01:09:09] - [01:09:12]

Jitesh Dugar: Now, with that, let's go ahead and execute this step.

[01:09:13] - [01:09:17]

Jitesh Dugar: Where we are now asking AI to classify, and you see what?

[01:09:17] - [01:09:21]

Jitesh Dugar: AI has classified it as a customer support-related email.

[01:09:22] - [01:09:36]

Jitesh Dugar: AI has classified it as a customer support email, and for people who are asking, can you only pin the trigger step, you can trigger... pin the action step also here. So there is a pin option here, and we will pin it.

[01:09:38] - [01:09:41]

Jitesh Dugar: I will repeat what I have done here. So...

[01:09:41] - [01:09:48]

Jitesh Dugar: What we've done is we've given a prompt to OpenAI Step, and we are telling it that you need to classify an incoming email

[01:09:48] - [01:09:52]

Jitesh Dugar: As being a customer support email or not customer support email?

[01:09:53] - [01:10:06]

Jitesh Dugar: And we are saying you should only reply with either CS or not CS. No other replies you're supposed to send. And then we would again say the content of the email that you're supposed to reply

[01:10:06] - [01:10:08]

Jitesh Dugar: This is here.

[01:10:08] - [01:10:10]

Jitesh Dugar: And the content of the email looks like this.

[01:10:11] - [01:10:15]

Jitesh Dugar: And for that, you simply... snippet means the email body, rather.

[01:10:15] - [01:10:17]

Jitesh Dugar: Email body, that's the snippet.

[01:10:18] - [01:10:26]

Jitesh Dugar: So you drag the subject from the left-hand side, put it in the subject field, you drag the snippet from the left-hand side, put it in the snippet field right here.

[01:10:28] - [01:10:36]

Jitesh Dugar: How do you know which is the subject in the email body? Subject already shows up as subject. Snippet show... the email body shows up as snippet.

[01:10:39] - [01:10:43]

Jitesh Dugar: So with that, you can execute the step, and you will get this kind of an output.

[01:10:44] - [01:10:47]

Jitesh Dugar: And this step has done the classification step for you.

[01:10:49] - [01:10:58]

Jitesh Dugar: Let us go ahead and rename it, and we'll call it... classify as CS or not.

[01:10:58] - [01:11:02]

Jitesh Dugar: Because this helps us classify whether it is customer support-related ticket or not.

[01:11:06] - [01:11:18]

Jitesh Dugar: So, we've just added a prompt here, and it gave us a message. It is CS. So, this step, just to summarize, this was the step where we used an LLM to classify an incoming email

[01:11:19] - [01:11:20]

Jitesh Dugar: whether it is CS or not.

[01:11:21] - [01:11:24]

Jitesh Dugar: And it replied with a CS for us.

[01:11:24] - [01:11:28]

Jitesh Dugar: And next step is where we will have to add a branching.

[01:11:29] - [01:11:31]

Jitesh Dugar: Where we have to add a branching.

[01:11:31] - [01:11:34]

Jitesh Dugar: And what should we use to create the branches?

[01:11:37] - [01:11:37]

Jitesh Dugar: Yes.

[01:11:38] - [01:11:43]

Jitesh Dugar: You can use a if, or you can use a switch. Both are okay.

[01:11:44] - [01:11:49]

Jitesh Dugar: So, let me use an if itself today. We'll use an IF step.

[01:11:51] - [01:11:57]

Jitesh Dugar: And we select an if step, and the condition that we are supposed to add for the if step

[01:11:57] - [01:12:01]

Jitesh Dugar: Will be, we have to check it against this value right here.

[01:12:01] - [01:12:06]

Jitesh Dugar: The value that we're getting in the text field, whether it is equal to CS or not.

[01:12:06] - [01:12:15]

Jitesh Dugar: So, we will drag this field under value 1, and we would check if this value is equal to CS.

[01:12:15] - [01:12:17]

Jitesh Dugar: That's what we are supposed to check.

[01:12:18] - [01:12:19]

Jitesh Dugar: And then?

[01:12:20] - [01:12:22]

Jitesh Dugar: We would execute the step.

[01:12:23] - [01:12:27]

Jitesh Dugar: If we execute this step, we are getting in the output

[01:12:28] - [01:12:33]

Jitesh Dugar: True branch, one item. False branch, nothing. False branch is empty.

[01:12:33] - [01:12:37]

Jitesh Dugar: That means it decided that for this given input that we had.

[01:12:38] - [01:12:40]

Jitesh Dugar: It is actually a customer support email.

[01:12:41] - [01:12:43]

Jitesh Dugar: That's why it went to the true branch.

[01:12:45] - [01:12:52]

Jitesh Dugar: Is it making sense, everyone, so far, on a scale of 1 to 10? Give me a score, please.

[01:12:52] - [01:12:54]

Jitesh Dugar: So I know how we are going.

[01:12:57] - [01:12:59]

Jitesh Dugar: Okay, perfect, okay.

[01:12:59] - [01:13:01]

Jitesh Dugar: No, it's just 1 to 10, not 100 or 1,000.

[01:13:10] - [01:13:12]

Jitesh Dugar: Okay, perfect. Thank you, everyone, for paying attention.

[01:13:14] - [01:13:28]

Jitesh Dugar: Okay, one of you said, can you go back on the if block? If block is this, the AI step, this is what it is. Now, remember, right, whatever we've done, you will have access to the workflow and all of that. You can execute it yourself later.

[01:13:28] - [01:13:36]

Jitesh Dugar: All of the content will be provided. For now, please pay full attention on the workbook, or the workflow that we are trying to do right now.

[01:13:38] - [01:13:39]

Jitesh Dugar: No.

[01:13:39] - [01:13:43]

Jitesh Dugar: Again, The most interesting aspect now, is...

[01:13:44] - [01:13:46]

Jitesh Dugar: If it is related to customer support.

[01:13:47] - [01:13:53]

Jitesh Dugar: If it is related to customer support, so this is the step that we've already done, we have to now use an AI agent.

[01:13:54] - [01:13:55]

Jitesh Dugar: So...

[01:13:55] - [01:14:02]

Jitesh Dugar: On the plus button here, when it is related to customer support, we have to add an AI agent.

[01:14:02] - [01:14:06]

Jitesh Dugar: So, on NA10, there is an option for you to select an AI agent.

[01:14:06] - [01:14:09]

Jitesh Dugar: Or, you can click on the AI step.

[01:14:09] - [01:14:12]

Jitesh Dugar: And then select an AI agent.

[01:14:13] - [01:14:19]

Jitesh Dugar: Select an AI agent. The moment you select an AI agent, you will be welcomed with a screen like this.

[01:14:23] - [01:14:24]

Jitesh Dugar: And...

[01:14:24] - [01:14:30]

Jitesh Dugar: On the screen, the first thing that it is asking is the source for the prompt, the user message.

[01:14:31] - [01:14:32]

Jitesh Dugar: Which means...

[01:14:32] - [01:14:42]

Jitesh Dugar: User message is the instruction that the AI agent takes, which means whenever you're talking to an AI agent, it will take a user message and it'll give you a response to the user back.

[01:14:43] - [01:14:45]

Jitesh Dugar: So that is what a user message means.

[01:14:46] - [01:14:47]

Jitesh Dugar: Nope.

[01:14:54] - [01:15:00]

Jitesh Dugar: Now, so, on this, By default, it would have selected Connected Chart Trigger Node.

[01:15:01] - [01:15:09]

Jitesh Dugar: But we do not use a chat trigger yet. We don't have a chat trigger at all. So we would say we would want to define the user message below.

[01:15:09] - [01:15:16]

Jitesh Dugar: We would want to define the user message below. Okay, people who are asking what is a user message, I will just quickly explain.

[01:15:16] - [01:15:18]

Jitesh Dugar: The concept of a user message.

[01:15:18] - [01:15:20]

Jitesh Dugar: Everyone.

[01:15:21] - [01:15:23]

Jitesh Dugar: In case of an AI agent.

[01:15:23] - [01:15:25]

Jitesh Dugar: There are 3 kinds of messages.

[01:15:27] - [01:15:31]

Jitesh Dugar: Or 3 kinds of prompts, rather. The three kinds of prompts are...

[01:15:31] - [01:15:33]

Jitesh Dugar: The first one is a system prompt.

[01:15:35] - [01:15:38]

Jitesh Dugar: The second one is a user prompt.

[01:15:39] - [01:15:41]

Jitesh Dugar: The third one is an assistant prompt.

[01:15:43] - [01:15:45]

Jitesh Dugar: or assistant message. It can be a...

[01:15:45] - [01:15:51]

Jitesh Dugar: Actually, assistant prompt, can... should ideally be called an assistant message.
Now.

[01:15:53] - [01:16:06]

Jitesh Dugar: Whatever is added in the guideline of the AI agent, that is a system prompt. System prompt is already built in within the AI agent itself, which means a lot of times, we cannot even look at the system prompt for AI agents, unless we have built it.

[01:16:07] - [01:16:10]

Jitesh Dugar: Only the person who's built the AI agent will be able to look at the system prompt.

[01:16:11] - [01:16:17]

Jitesh Dugar: User prompt is the message or the question that we are asking to the AI agent.

[01:16:17] - [01:16:23]

Jitesh Dugar: User prompt is the question that we are asking to the AI agent, so it is... this is a user prompt.

[01:16:24] - [01:16:29]

Jitesh Dugar: So, if you're asking it a question, what is the... What is the capital...

[01:16:30] - [01:16:35]

Jitesh Dugar: of the UK, for example, right? So this is the user prompt.

[01:16:36] - [01:16:37]

Jitesh Dugar: And?

[01:16:37] - [01:16:42]

Jitesh Dugar: AI Agent has a lot of guidelines on what it should do and all that, that's a system prompt.

[01:16:42] - [01:16:43]

Jitesh Dugar: And the third one?

[01:16:43] - [01:16:49]

Jitesh Dugar: Which is the... Assistant prompt is the reply that the AI agent gives.

[01:16:50] - [01:16:54]

Jitesh Dugar: So whatever the AI agent is giving in the reply, that is an assistant prompt.

[01:16:55] - [01:16:57]

Jitesh Dugar: Or assistant message.

[01:16:57] - [01:17:11]

Jitesh Dugar: So, these are the three types of prompts that you will deal with. But, as you can see, we can only control the system prompt and user prompt. Assistant prompt is something we don't control, because that's a reply that the AI agent gives.

[01:17:11] - [01:17:18]

Jitesh Dugar: So we... in a lot of cases, we will only deal with either a system prompt or a user prompt.

[01:17:19] - [01:17:20]

Jitesh Dugar: No.

[01:17:21] - [01:17:28]

Jitesh Dugar: With that, let's come back here. So we are saying the instruction that the AI agent should take, we will want to define it below.

[01:17:28] - [01:17:30]

Jitesh Dugar: And we will put it here.

[01:17:31] - [01:17:32]

Jitesh Dugar: And...

[01:17:32] - [01:17:37]

Jitesh Dugar: After that, there is an option for us to add a system prompt also, which I will show you later.

[01:17:37] - [01:17:42]

Jitesh Dugar: For now, in the user message, we will tell the AI agent

[01:17:42] - [01:17:47]

Jitesh Dugar: Everyone, please pay attention, this is going to be very interesting, and it could confuse you if you're not paying attention.

[01:17:48] - [01:17:48]

Jitesh Dugar: So...

[01:17:51] - [01:18:01]

Jitesh Dugar: Now we have to, in the user prompt, we have to tell it the email that it is supposed to process. The email that it is supposed to process. So, the email content

[01:18:02] - [01:18:04]

Jitesh Dugar: It's something that we have to give here.

[01:18:04] - [01:18:09]

Jitesh Dugar: Where will that email content come from? From the input on the left-hand side.

[01:18:09] - [01:18:12]

Jitesh Dugar: And that'll come from the Gmail trigger step.

[01:18:13] - [01:18:25]

Jitesh Dugar: Go to the Gmail trigger step, and the email snippet is already here, email subject is already here. And these are the two things that we need to give to the AI agent whenever we are communicating with it.

[01:18:26] - [01:18:33]

Jitesh Dugar: We would tell it the message that you're supposed to process is an email with this subject.

[01:18:33] - [01:18:35]

Jitesh Dugar: And an email with...

[01:18:35] - [01:18:46]

Jitesh Dugar: this body. So, because these are dynamic, we will switch to expression, and we will make the screen bigger by clicking on this arrow, and we will say.

[01:18:48] - [01:18:50]

Jitesh Dugar: Reply to this email.

[01:18:52] - [01:18:54]

Jitesh Dugar: Or process this email.

[01:18:55] - [01:19:01]

Jitesh Dugar: This is the instruction that we are giving to the AI agent. So we would say, the subject?

[01:19:02] - [01:19:05]

Jitesh Dugar: which will come from the Gmail trigger step, expand that.

[01:19:06] - [01:19:08]

Jitesh Dugar: And drag the subject, put it here.

[01:19:10] - [01:19:13]

Jitesh Dugar: And... email... Snippet.

[01:19:17] - [01:19:19]

Jitesh Dugar: Drag it and put it here.

[01:19:20] - [01:19:21]

Jitesh Dugar: That's what you do.

[01:19:22] - [01:19:25]

Jitesh Dugar: This is the email that the AI agent is supposed to process now.

[01:19:27] - [01:19:28]

Jitesh Dugar: No.

[01:19:28] - [01:19:33]

Jitesh Dugar: System prompt we still haven't given, which means we haven't told it what

[01:19:33] - [01:19:36]

Jitesh Dugar: are the guidelines that it should follow? We will come back to it.

[01:19:37] - [01:19:39]

Jitesh Dugar: With that, we will close this.

[01:19:40] - [01:19:45]

Jitesh Dugar: And do you remember the 3 things that the AI agent has? We discussed the first thing today.

[01:19:47] - [01:19:49]

Jitesh Dugar: A brain, a memory, and tools.

[01:19:49] - [01:19:57]

Jitesh Dugar: And on NATEN, AI Agent is now asking us to give it the brain The memory and the tools.

[01:19:58] - [01:20:02]

Jitesh Dugar: exactly like what we are expected to provide in an AI agent.

[01:20:02] - [01:20:07]

Jitesh Dugar: So... What do you think will a brain have, a chat model will have?

[01:20:08] - [01:20:12]

Jitesh Dugar: It will have all the LLMs. Yes, you guys are right, lovely.

[01:20:13] - [01:20:15]

Jitesh Dugar: Yes, brain will be that LLM.

[01:20:15] - [01:20:16]

Jitesh Dugar: So?

[01:20:17] - [01:20:20]

Jitesh Dugar: Click on the plus button, And...

[01:20:21] - [01:20:25]

Jitesh Dugar: There are all of these LLM options that you are able to choose from.

[01:20:25] - [01:20:33]

Jitesh Dugar: Whether you want to choose Anthropic, Google Gemini, Minimax, Moonshot, OpenAI, Quinn, all of these options are available, the most popular ones.

[01:20:34] - [01:20:38]

Jitesh Dugar: In case you want to use open source ones, there's OLama also that is supported.

[01:20:39] - [01:20:42]

Jitesh Dugar: For now, we'll use... OpenAI chat model.

[01:20:44] - [01:20:44]

Jitesh Dugar: None?

[01:20:45] - [01:20:49]

Jitesh Dugar: Just like how we used the previous OpenAI step, it is asking for the credentials.

[01:20:50] - [01:20:52]

Jitesh Dugar: It is asking you to choose the model.

[01:20:53] - [01:20:58]

Jitesh Dugar: Model... I personally don't prefer a GPT-5 mini, because I think it's too slow.

[01:20:58] - [01:21:05]

Jitesh Dugar: So, I would use maybe a other mini model, maybe a 5.4 mini, or a... I would use a 5.4 mini.

[01:21:06] - [01:21:07]

Jitesh Dugar: model.

[01:21:07] - [01:21:16]

Jitesh Dugar: And that's all you have to do on the chat model step. So you are telling that this agent needs to use this LLM, which is GPT 5.4 Vini.

[01:21:16] - [01:21:23]

Jitesh Dugar: as the LLM. And we have connected to our credentials. So these are the only two settings that you'll have to do on the chat model step.

[01:21:24] - [01:21:27]

Jitesh Dugar: With that, Your chat model is connected.

[01:21:29] - [01:21:30]

Jitesh Dugar: No.

[01:21:30] - [01:21:32]

Jitesh Dugar: Two important questions.

[01:21:32] - [01:21:35]

Jitesh Dugar: Rather, an important question for everyone.

[01:21:35] - [01:21:39]

Jitesh Dugar: What tools should the AI agent have access to?

[01:21:39] - [01:21:41]

Jitesh Dugar: this AI agent.

[01:21:42] - [01:21:44]

Jitesh Dugar: Should do all of these four things.

[01:21:45] - [01:21:49]

Jitesh Dugar: Yes, it should have access to Gmail, that's correct. What else?

[01:21:52] - [01:21:59]

Jitesh Dugar: Okay, so far, one... okay, a lot of other people have also answered. It should also have access to...

[01:21:59] - [01:22:01]

Jitesh Dugar: This given knowledge base.

[01:22:01] - [01:22:04]

Jitesh Dugar: It should have access to this given knowledge base.

[01:22:05] - [01:22:08]

Jitesh Dugar: So it should have an access to a knowledge base.

[01:22:10] - [01:22:18]

Jitesh Dugar: But... But the bakery guide, right, so let's call it this, the bakery guide, is what it is. Now.

[01:22:18] - [01:22:25]

Jitesh Dugar: But, do you remember this scenario that I told you earlier, Where...

[01:22:25] - [01:22:28]

Jitesh Dugar: Scenario 1 is where you give

[01:22:28] - [01:22:37]

Jitesh Dugar: the AI agent an ability to read the entire knowledge base in one single go, versus Scenario 2, where you give it extremely specific context.

[01:22:39] - [01:22:40]

Jitesh Dugar: So?

[01:22:40] - [01:22:51]

Jitesh Dugar: Is it fair to say that we really cannot give the entire knowledge base to the AI agent? We need to create some kind of a mechanism so that it does all of these things?

[01:22:51] - [01:22:57]

Jitesh Dugar: It should pick the relevant page that has the information. It should mix that with the question.

[01:22:57] - [01:23:00]

Jitesh Dugar: And then it should generate the answer.

[01:23:01] - [01:23:06]

Jitesh Dugar: That's what it should do. I've told it possibly twice or thrice, don't mind that, please.

[01:23:06] - [01:23:10]

Jitesh Dugar: Okay? I'm emphasizing on this because it's extremely important.

[01:23:11] - [01:23:12]

Jitesh Dugar: No.

[01:23:13] - [01:23:16]

Jitesh Dugar: We need to give it access to the knowledge base, but not directly.

[01:23:17] - [01:23:20]

Jitesh Dugar: We need to now convert it into a rack system.

[01:23:20] - [01:23:28]

Jitesh Dugar: this system that allows it to retrieve the right page, augment with the question, and generate the answer. That's what we're supposed to do.

[01:23:28] - [01:23:33]

Jitesh Dugar: And with that, we are about to get into a very interesting phase where we'll talk about RAG.

[01:23:34] - [01:23:35]

Jitesh Dugar: No.

[01:23:36] - [01:23:50]

Jitesh Dugar: How does this rack system work? Before that, we can give it access to the Gmail tool and all that. That's something that we will come back to. We'll come back to this workflow. But let me explain how this entire rack system works. Now.

[01:23:53] - [01:23:58]

Jitesh Dugar: Everyone, if you have any questions, curiosities, please keep that aside for some time.

[01:23:58] - [01:24:02]

Jitesh Dugar: Please keep that aside for some time. We will talk about a very important aspect called RAG.

[01:24:03] - [01:24:05]

Jitesh Dugar: And, let me explain how it works.

[01:24:06] - [01:24:09]

Jitesh Dugar: This knowledge-based document, so let's say you have a document.

[01:24:10] - [01:24:14]

Jitesh Dugar: what happens in document, and I promise you it'll be very useful.

[01:24:14] - [01:24:18]

Jitesh Dugar: You have this kind of a document. And the way this works is.

[01:24:19] - [01:24:23]

Jitesh Dugar: I actually think my drawing skills are not that great, so...

[01:24:23] - [01:24:26]

Jitesh Dugar: I have something that I could use.

[01:24:26] - [01:24:27]

Jitesh Dugar: To showcase it to you.

[01:24:29] - [01:24:33]

Jitesh Dugar: One minute, let me try this. I have something better to showcase.

[01:24:39] - [01:24:42]

Jitesh Dugar: Yes, I think this one should work.

[01:24:42] - [01:24:46]

Jitesh Dugar: Okay, I'll try something else rather than trying to draw it.

[01:24:46] - [01:24:48]

Jitesh Dugar: So, everyone.

[01:24:50] - [01:25:03]

Jitesh Dugar: This is the source document. This is the source document, and this is the exact document that we are talking about, okay? What we do in the process of RAG, there are two phases to it. You prepare the data first.

[01:25:03] - [01:25:03]

Jitesh Dugar: No.

[01:25:04] - [01:25:08]

Jitesh Dugar: When you try to prepare the data, what you do is, you...

[01:25:08] - [01:25:11]

Jitesh Dugar: This document goes through a process called chunking.

[01:25:12] - [01:25:16]

Jitesh Dugar: Chunking means breaking down that document into multiple parts.

[01:25:17] - [01:25:19]

Jitesh Dugar: Multiple small, small parts.

[01:25:19] - [01:25:20]

Jitesh Dugar: like this.

[01:25:21] - [01:25:33]

Jitesh Dugar: That document could be broken into 4, 5, 10, 20, 30, 40 chunks, whatever that is.

You can decide what it will work at, okay? So, chunking is like cutting that document into multiple parts.

[01:25:34] - [01:25:39]

Jitesh Dugar: So, example, we have broken this same document, Into all of these chunks.

[01:25:41] - [01:25:42]

Jitesh Dugar: Next.

[01:25:43] - [01:25:46]

Jitesh Dugar: What we do is... now, this is a very important aspect.

[01:25:47] - [01:25:51]

Jitesh Dugar: DONE... Chunks of the document, all the

[01:25:51] - [01:25:56]

Jitesh Dugar: Blocks of that document, or the chunks of the document, which is...

[01:25:57] - [01:26:02]

Jitesh Dugar: Which is in the form of text. Remember, these chunks are in the form of text.

[01:26:02] - [01:26:04]

Jitesh Dugar: This document is in the form of text.

[01:26:05] - [01:26:17]

Jitesh Dugar: Okay, I will repeat the whole thing if you're not paying attention, but let me tell you, this is very important. If you have any other questions, curiosity, keep it aside for the next 10 minutes only. 10-20 minutes per... 10-15 minutes only.

[01:26:18] - [01:26:20]

Jitesh Dugar: So I will repeat the whole thing again.

[01:26:21] - [01:26:23]

Jitesh Dugar: First thing, it starts with a source document.

[01:26:23] - [01:26:24]

Jitesh Dugar: Which is this.

[01:26:26] - [01:26:33]

Jitesh Dugar: It starts with a source talk. I'm only... I'm already repeating it, everyone. People who are asking me to repeat. I'm already repeating it. We're starting from scratch.

[01:26:34] - [01:26:37]

Jitesh Dugar: We start with the source document, which is this.

[01:26:39] - [01:26:45]

Jitesh Dugar: We go through the process where we chunk that document, which means break that document into multiple chunks, like this.

[01:26:48] - [01:26:53]

Jitesh Dugar: After that, What we do is... those chunks

[01:26:53] - [01:26:57]

Jitesh Dugar: Those chunks that we have, these are chunks in the form of text.

[01:26:58] - [01:27:00]

Jitesh Dugar: These chunks are in the form of text.

[01:27:01] - [01:27:04]

Jitesh Dugar: We... this text goes through a...

[01:27:04] - [01:27:17]

Jitesh Dugar: goes through a process, or a process called embedding, or rather an embedding model, that converts that text into a vector, that converts the text into a vector. Now, two important things that I've spoken about so far.

[01:27:17] - [01:27:21]

Jitesh Dugar: Embedding model and vector. Let me tell you what a vector is.

[01:27:21] - [01:27:29]

Jitesh Dugar: Vector is... Vector is a numerical representation of text.

[01:27:31] - [01:27:36]

Jitesh Dugar: Of text. And let me show you what a vector looks like. Let's say you have a text such as a cat.

[01:27:37] - [01:27:42]

Jitesh Dugar: Your text, such as cat, Goes through an embedding model.

[01:27:42] - [01:27:44]

Jitesh Dugar: It goes through an embedding model.

[01:27:52] - [01:27:53]

Jitesh Dugar: One minute...

[01:27:58] - [01:28:00]

Jitesh Dugar: Let's say you have a text cat.

[01:28:00] - [01:28:06]

Jitesh Dugar: This CAT will go through an embedding model, This is an embedding model.

[01:28:06] - [01:28:13]

Jitesh Dugar: And the duty of this embedding model is to convert this text into a set of numbers.

[01:28:13] - [01:28:16]

Jitesh Dugar: And these numbers look something like this.

[01:28:16] - [01:28:19]

Jitesh Dugar: They will look like maybe 0.4,

[01:28:20] - [01:28:32]

Jitesh Dugar: then a 0.7, then 0.1, then 0.6, then 0.2, I'm just giving random values right now. And then dot dot, dot dot, and a ton of these numbers.

[01:28:32] - [01:28:39]

Jitesh Dugar: And... This numerical representation of this text is called vectors.

[01:28:40] - [01:28:43]

Jitesh Dugar: The numerical representations of text is called vectors.

[01:28:45] - [01:28:55]

Jitesh Dugar: And you might be wondering, why are we even converting this to a vector? Because AI agents have a huge comfort level working with vectors rather than text.

[01:28:55] - [01:29:00]

Jitesh Dugar: So, AI agents, whenever they want to retrieve some information, Dey?

[01:29:01] - [01:29:11]

Jitesh Dugar: they are able to refer to numbers in a better way compared to this text. So, every single text is converted into a vector like this.

[01:29:11] - [01:29:18]

Jitesh Dugar: So, your chunk also goes through an embedding model, and it converts all of these texts

[01:29:18] - [01:29:20]

Jitesh Dugar: Into vector embeddings like this.

[01:29:22] - [01:29:26]

Jitesh Dugar: All of these text will convert into vector embeddings like this.

[01:29:26] - [01:29:33]

Jitesh Dugar: So, Chunk 1 will become vector embedding 1, Chung 2 will become vector embedding 2, Chung 3 will become vector embedding 3.

[01:29:34] - [01:29:39]

Jitesh Dugar: And finally, Finally, All of the vector

[01:29:39] - [01:29:46]

Jitesh Dugar: Embeddings that you have will be stored into a specialized database called a vector database.

[01:29:46] - [01:29:54]

Jitesh Dugar: These are specialized vector databases that will allow you to store all the embeddings and the chunks.

[01:29:56] - [01:29:58]

Jitesh Dugar: These are specialized databases.

[01:30:00] - [01:30:06]

Jitesh Dugar: Is it making sense, everyone? This is the whole data ingestion process, which is step one.

[01:30:08] - [01:30:10]

Jitesh Dugar: Yes, the last step is.

[01:30:10] - [01:30:16]

Jitesh Dugar: Once you have all of these vector embeddings, it gets stored into a specialized database called a vector database.

[01:30:16] - [01:30:22]

Jitesh Dugar: And there are a ton of vector database providers in the market. We will talk about those.

[01:30:23] - [01:30:30]

Jitesh Dugar: Yes, unlike regular databases, vector databases let you store all of these embeddings.

[01:30:32] - [01:30:38]

Jitesh Dugar: Yes, I did warn you, it might get a little complex, little overwhelming, and that's why I expect your full attention.

[01:30:40] - [01:30:41]

Jitesh Dugar: No.

[01:30:44] - [01:30:49]

Jitesh Dugar: Are there different embedding tools? There are different embedding models, not different embedding tools.

[01:30:49] - [01:30:51]

Jitesh Dugar: We will talk about that also.

[01:30:52] - [01:30:57]

Jitesh Dugar: This is not an application, this is an illustration that I built using Claude.

[01:30:58] - [01:30:59]

Jitesh Dugar: No.

[01:30:59] - [01:31:05]

Jitesh Dugar: This is just part one, everyone. This is just one part where your input was that source document.

[01:31:06] - [01:31:10]

Jitesh Dugar: Your input was that source document, And then...

[01:31:10] - [01:31:18]

Jitesh Dugar: that source document got converted into a vector database. So what does this vector database hold? It holds the

[01:31:18] - [01:31:24]

Jitesh Dugar: Embedded version, or the vector version, of this single document that you have.

[01:31:24] - [01:31:26]

Jitesh Dugar: This single document that you have.

[01:31:27] - [01:31:28]

Jitesh Dugar: That's what it does.

[01:31:29] - [01:31:30]

Jitesh Dugar: Now.

[01:31:33] - [01:31:40]

Jitesh Dugar: Now, if you're asking questions like, what's the benefits of chunking, we will talk about that in the next phase. So, this was where you prepared your data.

[01:31:40] - [01:31:42]

Jitesh Dugar: The vector database.

[01:31:42] - [01:31:45]

Jitesh Dugar: Now, second part is the most interesting part.

[01:31:45] - [01:31:49]

Jitesh Dugar: Where? How does it get answers to a question?

[01:31:50] - [01:32:01]

Jitesh Dugar: And remember, you can start relating it to the example that I shared earlier. Turning to the relevant page, reading the page along with the question, giving the answer in plain

[01:32:01] - [01:32:02]

Jitesh Dugar: English.

[01:32:03] - [01:32:09]

Jitesh Dugar: And we will go through a similar process now. I hope I have all your attention. So, the second part is where

[01:32:10] - [01:32:21]

Jitesh Dugar: you have to now answer a user's question. Let's say a user has given a question to the AI agent, and that question looks like this. What are the annual revenue figures for a company?

[01:32:22] - [01:32:23]

Jitesh Dugar: Next, what it does.

[01:32:24] - [01:32:25]

Jitesh Dugar: What happens is.

[01:32:26] - [01:32:34]

Jitesh Dugar: This question also goes through an embedding model, and you already know by now what is... what does the embedding model do? It converts text to a vector.

[01:32:35] - [01:32:38]

Jitesh Dugar: So, it converts this text

[01:32:38] - [01:32:52]

Jitesh Dugar: what are the annual revenue figures into a vector, query vector. So your question is now looking like numbers instead of text. Numbers that we cannot understand, but possibly these systems can understand.

[01:32:54] - [01:32:55]

Jitesh Dugar: Next.

[01:32:56] - [01:33:00]

Jitesh Dugar: Your question is now converted into a query vector using an embedding model.

[01:33:01] - [01:33:07]

Jitesh Dugar: And the next part is... Where there is a similarity search.

[01:33:07] - [01:33:14]

Jitesh Dugar: And where does that similarity search happen? It happens in the vector database. Now, remember, this is the step where you are

[01:33:14] - [01:33:16]

Jitesh Dugar: Turning the pages on that book.

[01:33:17] - [01:33:22]

Jitesh Dugar: Turning the pages on the book to find what is the closest looking

[01:33:22] - [01:33:33]

Jitesh Dugar: Stuff that may have. What is the page on this vector database, or what is the chunk inside this vector database that is likely to have an answer to this question?

[01:33:34] - [01:33:35]

Jitesh Dugar: I will repeat.

[01:33:36] - [01:33:41]

Jitesh Dugar: What happens here is, It will do a similarity search.

[01:33:42] - [01:33:54]

Jitesh Dugar: With the question vector among the vector database, it will try to identify which are the chunks that is likely to have an answer to the question that the user has asked.

[01:33:55] - [01:34:03]

Jitesh Dugar: And then... It will give us all the relevant chunks that is likely to have answers to that question.

[01:34:03] - [01:34:07]

Jitesh Dugar: Now, remember, this is like turning to that right page on that book.

[01:34:08] - [01:34:10]

Jitesh Dugar: Turning to that right page on the book.

[01:34:12] - [01:34:18]

Jitesh Dugar: So, you have all the relevant chunks from the vector database that you prepared in step number one.

[01:34:19] - [01:34:24]

Jitesh Dugar: And next, You augment that Those pages, or the chunks.

[01:34:24] - [01:34:31]

Jitesh Dugar: With your question, which means this question is now looked at with... along with the Chunk itself.

[01:34:32] - [01:34:36]

Jitesh Dugar: And then you have it pass through a large language model.

[01:34:37] - [01:34:42]

Jitesh Dugar: And to this model, instead of giving it the whole book, you will give it specific page numbers.

[01:34:42] - [01:34:45]

Jitesh Dugar: Specific pages or specific chunks.

[01:34:45] - [01:34:52]

Jitesh Dugar: And what will this large language model do? It will look at the augmented prompt.

[01:34:52] - [01:34:54]

Jitesh Dugar: And then it'll give you a final answer.

[01:34:55] - [01:34:59]

Jitesh Dugar: And that is the entire rack pipeline.

[01:34:59] - [01:35:03]

Jitesh Dugar: I am not sure if... This went over your head.

[01:35:03] - [01:35:04]

Jitesh Dugar: Okay?

[01:35:04] - [01:35:14]

Jitesh Dugar: And let me be honest, if you are introduced to the term RAG for the first time, this might be a bit overwhelming, but as you go ahead and learn about it, it's very interesting.

[01:35:15] - [01:35:17]

Jitesh Dugar: Now, on a scale of 1 to 10,

[01:35:17] - [01:35:20]

Jitesh Dugar: How much of it have you understood?

[01:35:22] - [01:35:34]

Jitesh Dugar: Okay, 1078783, okay? 96100, okay? 1078789, okay. Okay, 7, 5, okay. People who are saying below 5,

[01:35:34] - [01:35:38]

Jitesh Dugar: Okay, a lot of you have said 10, clear 10, that's good, okay?

[01:35:38] - [01:35:41]

Jitesh Dugar: And I'm happy to see that, but I will do a recap again.

[01:35:41] - [01:35:47]

Jitesh Dugar: But let me tell you one thing. If you're wondering, do we have to do all of these steps? Actually, no.

[01:35:47] - [01:35:50]

Jitesh Dugar: NAT has all of these things built in for you.

[01:35:51] - [01:36:01]

Jitesh Dugar: And it has all of these things built in for you. So you don't have to learn all of these, but it is better to understand it because sometimes you may have to deal with a few aspects of it.

[01:36:02] - [01:36:04]

Jitesh Dugar: So, I will do a quick recap, everyone.

[01:36:05] - [01:36:08]

Jitesh Dugar: First step is the data ingestion part.

[01:36:08] - [01:36:11]

Jitesh Dugar: Where you start with a source document.

[01:36:12] - [01:36:14]

Jitesh Dugar: Start with the source document, which is this.

[01:36:15] - [01:36:18]

Jitesh Dugar: This source document will be broken into all of these chunks.

[01:36:19] - [01:36:27]

Jitesh Dugar: Again, these are textual chunks. It goes to an embedding model that converts these chunks into a set of numbers.

[01:36:27] - [01:36:29]

Jitesh Dugar: Numbers will look like this.

[01:36:31] - [01:36:32]

Jitesh Dugar: And...

[01:36:32] - [01:36:40]

Jitesh Dugar: These numbers are then stored into the vector database. That's step number one, where you are preparing a vector database that will be used in step number two.

[01:36:42] - [01:36:50]

Jitesh Dugar: And step number two is where you have to answer the user's question. And you can always relate to the same three steps here.

[01:36:50] - [01:36:55]

Jitesh Dugar: Turning to the right page, mixing up that with the question, and getting the answer.

[01:36:55] - [01:36:58]

Jitesh Dugar: So, starts with the user's question.

[01:36:58] - [01:37:07]

Jitesh Dugar: It goes through an embedding model where the text is converted into a vector, which means the user's question is now looking like this. This is the user's question now.

[01:37:09] - [01:37:09]

Jitesh Dugar: And...

[01:37:10] - [01:37:19]

Jitesh Dugar: Then there is a similarity search, where you do a search on that vectors database to identify which is the closest chunk

[01:37:19] - [01:37:24]

Jitesh Dugar: Inside of this vector database that is likely to have an answer to this question.

[01:37:27] - [01:37:32]

Jitesh Dugar: And it will identify those chunks, Those specific page numbers.

[01:37:32] - [01:37:40]

Jitesh Dugar: it will... Augment that with the question, which means these chunks, along with the question, will now go to

[01:37:40] - [01:37:47]

Jitesh Dugar: A large language model, That is great at guessing, but now it has better context to guess.

[01:37:47] - [01:37:48]

Jitesh Dugar: So?

[01:37:49] - [01:37:53]

Jitesh Dugar: And that large language model will give you the final answer.

[01:37:55] - [01:37:59]

Jitesh Dugar: With context from the vector database, which is nothing but this document.

[01:38:01] - [01:38:07]

Jitesh Dugar: Is it making sense? Vectors are not binary numbers. Vectors are altogether different kind of numbers.

[01:38:08] - [01:38:12]

Jitesh Dugar: Okay? Now... No.

[01:38:15] - [01:38:21]

Jitesh Dugar: And as we go further, I will introduce you to some terminologies also, okay? Now, okay.

[01:38:24] - [01:38:25]

Jitesh Dugar: Everyone.

[01:38:26] - [01:38:42]

Jitesh Dugar: Even if you've not understood, I can see one or two people who've said you've not understood. Now, I will explain that again when we implement it. And when I implement it, you'll start to realize it's actually not so complex to implement it. So, let's go ahead and start implementing it. So.

[01:38:42] - [01:38:51]

Jitesh Dugar: The first step, everyone, for us is data preparation. Data preparation, where we will feed that source document and convert that into a vector database.

[01:38:51] - [01:38:57]

Jitesh Dugar: So... so our step will look like this. This is step number one. So, how do we do that on Ann?

[01:38:59] - [01:39:07]

Jitesh Dugar: And for that, we need to choose a vector database. We need to first decide what vector database to choose.

[01:39:07] - [01:39:17]

Jitesh Dugar: And for that, the one that we will choose here is Pinecone. The website is pinecone.io. It is one of the most trusted and popular vector databases right now.

[01:39:18] - [01:39:36]

Jitesh Dugar: You are free to choose any other vector databases also. There are a ton of other options as well. I could have chosen to use N80's vector database also. NA10 also has a vector database, but it's not at a scale where you can use it for heavy use cases. But Pinecone can be used for a lot of use cases.

[01:39:37] - [01:39:39]

Jitesh Dugar: And it has a very generous free plan.

[01:39:40] - [01:39:42]

Jitesh Dugar: So let's click on login.

[01:39:46] - [01:39:48]

Jitesh Dugar: You can sign up for a free account.

[01:39:49] - [01:39:52]

Jitesh Dugar: And after you sign up for a free account, you'll come to a page like this.

[01:39:52] - [01:39:57]

Jitesh Dugar: And... Over here, you can create your vector databases.

[01:39:57] - [01:39:57]

Jitesh Dugar: No.

[01:39:58] - [01:40:02]

Jitesh Dugar: We have to first create a vector database on Pinecone.

[01:40:02] - [01:40:09]

Jitesh Dugar: the vector database where we will store the knowledge base. So, we will click on database on the left-hand side.

[01:40:09] - [01:40:13]

Jitesh Dugar: It automatically takes you to indexes where you create your database.

[01:40:13] - [01:40:17]

Jitesh Dugar: So you click on Create Index, which is like creating that vector database.

[01:40:18] - [01:40:23]

Jitesh Dugar: So, we will just give it a name, we'll call it... Bake house.

[01:40:25] - [01:40:26]

Jitesh Dugar: knowledge base.

[01:40:26] - [01:40:28]

Jitesh Dugar: And then it is asking.

[01:40:28] - [01:40:34]

Jitesh Dugar: Where would you like to host it? What configuration you want to use? All of these settings can remain by default.

[01:40:35] - [01:40:45]

Jitesh Dugar: cloud provider, AWS, which location you want to host it and all that, that is all up to you, okay? Now, one thing that I want all of you to remember, there is something called dimension.

[01:40:45] - [01:40:50]

Jitesh Dugar: And dimension is set to 1024, and I will make a note of it.

[01:40:51] - [01:40:57]

Jitesh Dugar: vector database, Has something called... dimension.

[01:40:58] - [01:41:03]

Jitesh Dugar: And the dimension that we are setting right now is 1024, is the dimension that we are setting.

[01:41:03] - [01:41:05]

Jitesh Dugar: And let's create the index.

[01:41:09] - [01:41:12]

Jitesh Dugar: Now, let me tell you what a dimension is.

[01:41:16] - [01:41:16]

Jitesh Dugar: Yeah.

[01:41:17] - [01:41:22]

Jitesh Dugar: So... You remember I told you that a text or a word like cat

[01:41:22] - [01:41:28]

Jitesh Dugar: uses an embedding model... using an embedding model, it gets converted into a set of vectors.

[01:41:29] - [01:41:30]

Jitesh Dugar: And?

[01:41:30] - [01:41:33]

Jitesh Dugar: These vectors have all of these numbers.

[01:41:33] - [01:41:41]

Jitesh Dugar: the count of the numbers here, and I give you a dot dot dot here. So, here, there is 1, 2, 3, 4, 5.

[01:41:41] - [01:41:50]

Jitesh Dugar: That is 5 dimensions. Imagine if you had 1024 combinations of such numbers. That is what a dimension means.

[01:41:52] - [01:42:02]

Jitesh Dugar: And dimension is nothing but... I mean, right now, we technically look at two-dimension, or three-dimension, or four dimensions. AI agents work in 1024 dimensions.

[01:42:02] - [01:42:08]

Jitesh Dugar: So they have a space where they place all of these texts one after the other.

[01:42:10] - [01:42:14]

Jitesh Dugar: These texts are placed one after the other, where you look... you can look at each of the texts.

[01:42:15] - [01:42:22]

Jitesh Dugar: And they are added in the form of dimension. So, all the count of numbers that you have, these are dimensions.

[01:42:23] - [01:42:29]

Jitesh Dugar: So this means there would be 1,024 combinations of these numbers to form that vector.

[01:42:33] - [01:42:36]

Jitesh Dugar: Yes, why we chose 1024, I will explain that also.

[01:42:38] - [01:42:39]

Jitesh Dugar: So for now.

[01:42:39] - [01:42:46]

Jitesh Dugar: The database is being prepared right now, and the database is ready. The vector database is ready right now.

[01:42:47] - [01:42:50]

Jitesh Dugar: And it's saying it does not have any information yet.

[01:42:51] - [01:42:53]

Jitesh Dugar: It does not have any information yet.

[01:42:54] - [01:42:58]

Jitesh Dugar: We are supposed to load information into this vector database. Now.

[01:43:02] - [01:43:06]

Jitesh Dugar: Okay, do we need a 5-minute break, everyone?

[01:43:10] - [01:43:13]

Jitesh Dugar: Okay, let's take a 5-minute break, I'll start a timer.

[01:43:13] - [01:43:15]

Jitesh Dugar: And I'll see you after the break.

[01:48:28] - [01:48:30]

Jitesh Dugar: Alright, are we back, everyone?

[01:48:36] - [01:48:39]

Jitesh Dugar: Okay, perfect. Perfect, thank you. With that.

[01:48:40] - [01:48:50]

Jitesh Dugar: I got a message from a few people, because we wanted to understand dimensions better. I thought maybe I will explain all the terminologies, one after the other, giving you some

[01:48:51] - [01:48:57]

Jitesh Dugar: instances or ideas as well. So, and this handbook that I have, I will send it across to you.

[01:48:58] - [01:49:07]

Jitesh Dugar: So, the first thing, we already know what a RAG is, I'll not get into that. We've seen these, phases also, how the data preparation stage works, and how the question answering works.

[01:49:08] - [01:49:09]

Jitesh Dugar: After that.

[01:49:10] - [01:49:17]

Jitesh Dugar: The chunking aspect. Let us try to recap on the, terminologies and how we go about each of those.

[01:49:17] - [01:49:32]

Jitesh Dugar: Chunking means breaking that large document into smaller, manageable pieces. And I got a question from some people, why are you even doing chunking? Because you need to create those different page numbers, and you can use those specific chunks

[01:49:32] - [01:49:40]

Jitesh Dugar: to send it as a context to the AI agent, or rather the LLM, rather, not AI agent, LLM, to get the right answer.

[01:49:41] - [01:49:53]

Jitesh Dugar: So, context, chunky means, let's say if you have a document like this, that has... maybe that mentions chapter 1, Introduction to Python, Python is a programming language, it was created by this, this, this.

[01:49:53] - [01:49:55]

Jitesh Dugar: So what happens is chunk 1 will look like...

[01:49:56] - [01:50:03]

Jitesh Dugar: You would say, I would like to break it into usable chunks, so I would say this would become chunk 1 for me.

[01:50:04] - [01:50:08]

Jitesh Dugar: And Python is used for web development, data science, and automation. That'll become Chung 2.

[01:50:09] - [01:50:15]

Jitesh Dugar: And then this will become Chung 3. So I break it into multiple sizable chunks.

[01:50:17] - [01:50:25]

Jitesh Dugar: And why do we chunk? Because AI models have a limit on how much text they can process at once. Smaller chunks are easier to search, and it works faster also.

[01:50:26] - [01:50:30]

Jitesh Dugar: Again, there is a concept of chunk size.

[01:50:31] - [01:50:39]

Jitesh Dugar: That means... When you're trying to break down that document, when you try to break down that document, Then...

[01:50:39] - [01:50:49]

Jitesh Dugar: what... I mean, how should you break it down? Should you try to break it by maybe 100, 200 characters, maybe 500 to 1,000 characters? What is better?

[01:50:49] - [01:50:50]

Jitesh Dugar: No.

[01:50:51] - [01:50:54]

Jitesh Dugar: If you try to break it into smaller chunks.

[01:50:54] - [01:51:01]

Jitesh Dugar: It works well, right? It works with high precision, but the context is low, because every chunk is very small.

[01:51:02] - [01:51:09]

Jitesh Dugar: If you try to break it into large, large chunks, which is great, the precision is low, but the context is high.

[01:51:09] - [01:51:11]

Jitesh Dugar: So it means it'll work slower.

[01:51:11] - [01:51:26]

Jitesh Dugar: But it will try to dilute the answer. So, suggestion is to start with something between 500 to 1,000 characters to start with. And based on your use case, you can decide whether you want to use a smaller chunk size, a larger chunk size, that's up to you.

[01:51:27] - [01:51:32]

Jitesh Dugar: No, dimension is not chunk size, that's a different term. Dimension and chunk size are different.

[01:51:32] - [01:51:37]

Jitesh Dugar: So, chunk size means how big each piece should be. In this.

[01:51:37] - [01:51:42]

Jitesh Dugar: How much should be the size of chunk 1, and how much should be the size of chunk 2 and chunk 3?

[01:51:42] - [01:51:49]

Jitesh Dugar: That's how chunk size you define. Now, the next important term, which is my favorite, Text splitters.

[01:51:49] - [01:51:53]

Jitesh Dugar: Okay? Now... Everyone, please pay attention.

[01:51:53] - [01:51:59]

Jitesh Dugar: So... Option that you have is Character Text Splitter.

[01:51:59] - [01:52:13]

Jitesh Dugar: Which means, when you try to break the document into multiple chunks, how do you go about breaking it? So, if you say, I want the split to happen every 50 characters, and if this was your original text.

[01:52:14] - [01:52:16]

Jitesh Dugar: It would have been broken into.

[01:52:16] - [01:52:21]

Jitesh Dugar: This part, Python is a programming language, it is easy to... which is exactly 50 characters.

[01:52:22] - [01:52:24]

Jitesh Dugar: And the second chunk would be this.

[01:52:25] - [01:52:34]

Jitesh Dugar: The second chunk would say, learn Python is versatile. But the challenge with that is, it cut middle of the sentence, which is not right.

[01:52:35] - [01:52:39]

Jitesh Dugar: So, if it cuts middle of the sentence, it is not right.

[01:52:40] - [01:52:45]

Jitesh Dugar: And that's why... There is another way of splitting the text.

[01:52:45] - [01:52:54]

Jitesh Dugar: And which is what majority of the use cases use. More than 90% of the use cases use not a character text splitter, but

[01:52:54] - [01:53:04]

Jitesh Dugar: Recursive character text splitter, and this is a very interesting concept. What it does is, instead of breaking by fixed number of characters.

[01:53:04] - [01:53:13]

Jitesh Dugar: Instead of breaking by the fixed number of characters, it will try to split at natural boundaries. It could be paragraphs, then sentences, then words.

[01:53:13] - [01:53:18]

Jitesh Dugar: And then if that also doesn't work out, then it will go into raw character split.

[01:53:18] - [01:53:25]

Jitesh Dugar: So, this is how the priority looks like. It will try to break it by double-line breaks, paragraphs first.

[01:53:25] - [01:53:38]

Jitesh Dugar: If you are able to get to the number of characters, then if you're not able to do it, then break it by the single line breaks, otherwise sentences, then words, then by individual characters. So this is a recursive character text splitter.

[01:53:38] - [01:53:46]

Jitesh Dugar: And... And a practical example which... where it will make extremely good sense is this.

[01:53:46] - [01:53:47]

Jitesh Dugar: So if you're...

[01:53:47] - [01:53:56]

Jitesh Dugar: document looks like this, where it says, Chapter 1, Introduction to Python. Python is a high-level programming language, and all of these things.

[01:53:56] - [01:54:16]

Jitesh Dugar: you're... and you're saying, I want to break it using recursive character text splitter, this text splitter mechanism, the chunking mechanism, okay? So what it will do is, with max 100 characters, what it'll do is, chunk 1 will look like this. Chapter 1, Introduction to Python. Python is a high-level programming language.

[01:54:16] - [01:54:26]

Jitesh Dugar: And Chung 2 will look like it was created by Gaidu Van Rosam in 1991. And Chung 3 will look like Python is known for its simplicity. Many beginners start with Python.

[01:54:27] - [01:54:34]

Jitesh Dugar: Now, imagine, this is so much better compared to the previous one, where it was breaking in the middle of the sentence.

[01:54:35] - [01:54:39]

Jitesh Dugar: And each chunk is complete and making sense on its own.

[01:54:39] - [01:54:44]

Jitesh Dugar: And this mechanism of chunking is called recursive character text splitter.

[01:54:45] - [01:54:50]

Jitesh Dugar: And remember, we don't have to do all of these, we will be able to configure this within NATAM.

[01:54:50] - [01:54:52]

Jitesh Dugar: I'm just telling you how it works.

[01:54:53] - [01:55:02]

Jitesh Dugar: Everyone good until now, you understand the chunking approaches, the fixed character text splitter approach, and the recursive character text splitter approach.

[01:55:04] - [01:55:11]

Jitesh Dugar: Why does the first one even exist? I wonder that. That's an easy one to implement, but...

[01:55:12] - [01:55:15]

Jitesh Dugar: it is better to use a recursive character text splitter. So.

[01:55:16] - [01:55:19]

Jitesh Dugar: This is an example for it. Now...

[01:55:19] - [01:55:30]

Jitesh Dugar: And these are all theoretical stuff. I will provide you this document anyways, you can go through it anytime, refer to it, you will always have access to it. And this has all the understanding of how you go about it.

[01:55:31] - [01:55:37]

Jitesh Dugar: Again, the next concept is... The next concept is embeddings.

[01:55:37] - [01:55:43]

Jitesh Dugar: And I have already told you, embeddings is the process of converting text to numbers.

[01:55:43] - [01:55:53]

Jitesh Dugar: So, your text, such as cat will be converted into a set of numbers like this. 0.20.8, 0.3, 0.9, and dot dot dot after that.

[01:55:53] - [01:55:54]

Jitesh Dugar: Okay?

[01:55:54] - [01:55:58]

Jitesh Dugar: And kitten will be converted to something like this.

[01:55:58] - [01:55:59]

Jitesh Dugar: No.

[01:56:00] - [01:56:01]

Jitesh Dugar: And...

[01:56:02] - [01:56:09]

Jitesh Dugar: And... and these are vectors, right? So, this text is converted into this set of numbers, which are technically vectors.

[01:56:09] - [01:56:10]

Jitesh Dugar: No.

[01:56:10] - [01:56:15]

Jitesh Dugar: One more thing to also remember, if you're... if you're trying to look at this.

[01:56:15] - [01:56:20]

Jitesh Dugar: The way the vectors database stores all of these vectors are.

[01:56:21] - [01:56:28]

Jitesh Dugar: They store it based on how similar each words are. For example, cat and kitten are very similar to each other.

[01:56:28] - [01:56:29]

Jitesh Dugar: So...

[01:56:30] - [01:56:42]

Jitesh Dugar: The numbers that you have for cat and for kitten, they are very similar. So here you can see it's 0.2, and this is 0.3. This is 0.8, this is 0.7, very close to each other. So in the vector

[01:56:42] - [01:56:46]

Jitesh Dugar: vector database world that has a set of so many words.

[01:56:46] - [01:56:49]

Jitesh Dugar: These will be stored very close to each other.

[01:56:49] - [01:56:56]

Jitesh Dugar: Now, if someone asked me what does a vector stored look like, I actually found out something.

[01:56:56] - [01:57:05]

Jitesh Dugar: Where this is what a vector store looks like. It has all of these numbers, all of these words placed in multi-dimensional space.

[01:57:05] - [01:57:14]

Jitesh Dugar: multi-dimensional space. We can only look through three dimensions, or maybe slightly more than that, but the way AI agents work is they work in multi-dimensions.

[01:57:14] - [01:57:25]

Jitesh Dugar: So this is a representation of how each word is stored. So bread is stored like this, you can go ahead and look at other words, and this is how it works. We really can... as humans, we cannot understand it.

[01:57:25] - [01:57:30]

Jitesh Dugar: So, if you're talking about dimension, this is 1024 dimensions.

[01:57:31] - [01:57:33]

Jitesh Dugar: Impossible for us to even visualize.

[01:57:37] - [01:57:43]

Jitesh Dugar: And these... all of these words are stored inside of these vectors.

[01:57:43] - [01:57:48]

Jitesh Dugar: Now, what are dimensions? Dimensions are the length of that number list.

[01:57:48] - [01:57:55]

Jitesh Dugar: The number of numbers that you have inside the vector representation, the length of that is a dimension.

[01:57:56] - [01:58:02]

Jitesh Dugar: And... How do you decide the dimension? There is a small cheat sheet here.

[01:58:02] - [01:58:13]

Jitesh Dugar: If you're looking at faster speed, go for lower dimensions. Go for 384 dimension, it'll work really fast. And it consumes less storage, but it has lower accuracy.

[01:58:14] - [01:58:26]

Jitesh Dugar: So, imagine, think of it like a video quality. 480p video quality, you will get the idea, but you will miss the detail. It consumes less space, it's easier to manage, cheaper also, but you miss the detail.

[01:58:26] - [01:58:30]

Jitesh Dugar: Medium is where you decide a dimension, where...

[01:58:30] - [01:58:33]

Jitesh Dugar: The speed is good, the storage is also good.

[01:58:33] - [01:58:38]

Jitesh Dugar: accuracy is also decent. So, 768 or something around 1024, they're all good enough.

[01:58:40] - [01:58:43]

Jitesh Dugar: Large means your dimensions are very high.

[01:58:43] - [01:58:51]

Jitesh Dugar: That means your accuracy level is the best. You have the best level of accuracy in terms of getting responses, but it is slower as well.

[01:58:51] - [01:59:00]

Jitesh Dugar: So you can decide if you are working with legal tech businesses or healthcare businesses where accuracy is extremely important, use a large dimension.

[01:59:01] - [01:59:05]

Jitesh Dugar: Use a larger dimension. And it also consumes more token as well.

[01:59:05] - [01:59:07]

Jitesh Dugar: for larger dimensions.

[01:59:08] - [01:59:14]

Jitesh Dugar: Then, VectorStore is a specialized databases that help you store those embeddings, or rather vectors.

[01:59:15] - [01:59:18]

Jitesh Dugar: That's how... that's what a vector store is.

[01:59:18] - [01:59:22]

Jitesh Dugar: So these are all the terminologies, the theoretical stuff.

[01:59:25] - [01:59:26]

Jitesh Dugar: Next.

[01:59:26] - [01:59:43]

Jitesh Dugar: Chunk Overlap, okay? Very interesting concept. When I started this session today, I gave you a small recap from what we discussed yesterday, okay? And that's how chunk

overlap also works. In TV shows, you have a recap of the previous episode, before starting the new one.

[01:59:43] - [01:59:51]

Jitesh Dugar: And why is that important? Because if you don't have that, that, you possibly miss the context. So overlap is that recap.

[01:59:51] - [01:59:53]

Jitesh Dugar: Let me show you how overlap works.

[01:59:54] - [01:59:59]

Jitesh Dugar: Let's say if your main sentence looks like this, the company grew from 10 employees to 100 employees.

[02:00:01] - [02:00:07]

Jitesh Dugar: This rapid growth requires new office space. So, chunk one will be, the company grew from 10 employees to 100 employees.

[02:00:07] - [02:00:20]

Jitesh Dugar: And there will be an overlap of 200 employees, which means this 200 employees, this part, will be appearing in this chunk 1 also, and in chunk 2 also, which means it's overlapping between both of these chunks.

[02:00:22] - [02:00:27]

Jitesh Dugar: So with that, the other chunk has a context of what is being talked about.

[02:00:27] - [02:00:29]

Jitesh Dugar: It is less likely to miss context.

[02:00:30] - [02:00:37]

Jitesh Dugar: And usually, in practice, You should remember that about a 10-20% overlap is good enough.

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Jitesh Dugar: Don't try to do too much, don't try to do too less.

[02:00:40] - [02:00:43]

Jitesh Dugar: 10-20% overlap is recommended.

[02:00:45] - [02:00:53]

Jitesh Dugar: With that, these are all the terminologies that we will be going across, and finally, I've put a cheat sheet on top on what you should be using. The settings that

[02:00:53] - [02:00:59]

Jitesh Dugar: we will be possibly using when we build our rack system, or the rack pipeline.

[02:01:02] - [02:01:05]

Jitesh Dugar: Everyone good with this? You want to see it in practice now?

[02:01:05] - [02:01:13]

Jitesh Dugar: We want to see the whole thing in practice. I hope you're all excited to see that. I am very excited to show it to all of you.

[02:01:14] - [02:01:16]

Jitesh Dugar: Yes, okay. With that...

[02:01:17] - [02:01:35]

Jitesh Dugar: Again, now, the first step for us is we have to do the data preparation part, where we will use this source document, which is nothing but this document, and we will load it inside a vector database. And we have already prepared our vector database inside Pinecone.

[02:01:35] - [02:01:39]

Jitesh Dugar: This is the pinecone, where we will load that document.

[02:01:40] - [02:01:40]

Jitesh Dugar: Now...

[02:01:41] - [02:01:50]

Jitesh Dugar: Let me show you how you do that. We will do it from within NA10, and for that, the first step is, we have to find a way to

[02:01:50] - [02:01:53]

Jitesh Dugar: Upload that document onto an NATN workflow.

[02:01:53] - [02:01:57]

Jitesh Dugar: And for that, I will add a node here.

[02:01:58] - [02:01:59]

Jitesh Dugar: I will add a chat node.

[02:02:00] - [02:02:06]

Jitesh Dugar: I will search for chat node, and I will use a trigger that says, on new chat event.

[02:02:06] - [02:02:08]

Jitesh Dugar: And I will simply close it.

[02:02:08] - [02:02:10]

Jitesh Dugar: With that, I will get something like this.

[02:02:10] - [02:02:14]

Jitesh Dugar: Which means this allows me to send a chat message onto this workflow.

[02:02:15] - [02:02:16]

Jitesh Dugar: No.

[02:02:16] - [02:02:20]

Jitesh Dugar: Click on Open Chat, you have a chat message bar at the bottom.

[02:02:20] - [02:02:23]

Jitesh Dugar: Where you can send a message. Now, everyone, please pay attention.

[02:02:23] - [02:02:26]

Jitesh Dugar: Our intention here is to send

[02:02:26] - [02:02:34]

Jitesh Dugar: this document, this document, onto this N10 workflow. But unfortunately, we do not have a way to attach a document.

[02:02:34] - [02:02:37]

Jitesh Dugar: So... To do that.

[02:02:37] - [02:02:40]

Jitesh Dugar: Go inside your chart, take a step.

[02:02:41] - [02:02:46]

Jitesh Dugar: Click on Options, Add Field, and enable this setting that says Allow File Uploads.

[02:02:46] - [02:02:52]

Jitesh Dugar: Why are we doing all of this? Because we want to... a way to load the file into this Ann workflow.

[02:02:52] - [02:02:56]

Jitesh Dugar: So we select Allow File Upload, and we enable it. And we close it.

[02:02:57] - [02:03:01]

Jitesh Dugar: Now, as we do that, you can see here, it has a way for us to upload a file.

[02:03:02] - [02:03:04]

Jitesh Dugar: And we can now click on it.

[02:03:05] - [02:03:07]

Jitesh Dugar: And select the file that we have.

[02:03:08] - [02:03:10]

Jitesh Dugar: Let me look for that file.

[02:03:16] - [02:03:20]

Jitesh Dugar: I have just uploaded the file, and I'll just say my

[02:03:21] - [02:03:24]

Jitesh Dugar: knowledge base, and I would send the message.

[02:03:24] - [02:03:26]

Jitesh Dugar: the moment I send the message.

[02:03:27] - [02:03:31]

Jitesh Dugar: it got executed with a green tick mark. If you... if I double-click on it.

[02:03:32] - [02:03:38]

Jitesh Dugar: If we double-click on it, It just shows the document in front of us.

[02:03:38] - [02:03:44]

Jitesh Dugar: All that we have done so far is we have uploaded that document inside this NA10 workflow.

[02:03:44] - [02:03:47]

Jitesh Dugar: So we have this inside of our Ann10 workflow.

[02:03:48] - [02:03:53]

Jitesh Dugar: Everyone, people who ask me to repeat, I will just show it again, but please pay attention.

[02:03:53] - [02:03:58]

Jitesh Dugar: I am right now clicking on Add Node. I will select a chat trigger.

[02:03:59] - [02:04:04]

Jitesh Dugar: I would say on new chat event, and I would go to Option, and I'll say Allow File Upload, and enable it.

[02:04:05] - [02:04:07]

Jitesh Dugar: And after that, I have a way to...

[02:04:09] - [02:04:11]

Jitesh Dugar: I have a way to upload a file.

[02:04:12] - [02:04:16]

Jitesh Dugar: Upload this same file that we were using, and we would just upload it.

[02:04:16] - [02:04:19]

Jitesh Dugar: And with that, the file has been uploaded.

[02:04:19] - [02:04:23]

Jitesh Dugar: And we have the file inside of this Ann workflow.

[02:04:24] - [02:04:25]

Jitesh Dugar: No.

[02:04:25] - [02:04:32]

Jitesh Dugar: Now, what are we supposed to do next? We are supposed to insert this, Into the vector store.

[02:04:32] - [02:04:36]

Jitesh Dugar: So, click on the plus button, And select.

[02:04:36] - [02:04:37]

Jitesh Dugar: Pinecone.

[02:04:37] - [02:04:40]

Jitesh Dugar: Because we want to insert that into a pinecone vector store.

[02:04:41] - [02:04:46]

Jitesh Dugar: Select Pinecone Vector Store, And the action that we have here is.

[02:04:46] - [02:04:54]

Jitesh Dugar: add documents to a vector store. Do you all agree that we now have a document, and we have to load that into a vector store?

[02:04:54] - [02:04:57]

Jitesh Dugar: So that's the most natural action that we can select.

[02:04:58] - [02:05:02]

Jitesh Dugar: So we uploaded a document, and we want to add it to a vector store. Select that.

[02:05:02] - [02:05:09]

Jitesh Dugar: And the first option that it is asking is, give me an ability to connect to your pinecone.

[02:05:09] - [02:05:10]

Jitesh Dugar: So?

[02:05:10] - [02:05:15]

Jitesh Dugar: Click on Connect to Pinecone. It is asking for the API key.

[02:05:15] - [02:05:18]

Jitesh Dugar: which we will get from Pinecone.

[02:05:18] - [02:05:23]

Jitesh Dugar: Inside Pinecone, you have to click on API Keys on the left-hand side.

[02:05:24] - [02:05:27]

Jitesh Dugar: Click on New API Key, you can give it any name.

[02:05:29] - [02:05:31]

Jitesh Dugar: Give it any name, click on Create Key.

[02:05:31] - [02:05:40]

Jitesh Dugar: This is your secret key for Pinecone. Copy that. This process is very similar as what you did for OpenAI also.

[02:05:40] - [02:05:44]

Jitesh Dugar: Come back here, paste the API key here, and save it.

[02:05:45] - [02:05:52]

Jitesh Dugar: With that, your API key is now added, and NATM now has an ability to use your Pinecone account.

[02:05:53] - [02:05:54]

Jitesh Dugar: And now it is asking.

[02:05:55] - [02:06:03]

Jitesh Dugar: which Pinecone index or the database that you want me to insert into? We already created the database, which was Bakehouse Knowledge Base.

[02:06:04] - [02:06:11]

Jitesh Dugar: So, let us open that on the left-hand side. This is the one where we need the vectors to show up.

[02:06:13] - [02:06:25]

Jitesh Dugar: Let's come back here, and all of these settings... embedding batch size means how many documents would you like to insert in one go? Right now, one is enough. 200 is default option, but we are only dealing with one right now.

[02:06:26] - [02:06:28]

Jitesh Dugar: After that, there are two options here.

[02:06:29] - [02:06:30]

Jitesh Dugar: Now, please pay attention.

[02:06:31] - [02:06:38]

Jitesh Dugar: When you're adding this, the first option that it is giving us is, it is asking us to choose the embedding model.

[02:06:39] - [02:06:49]

Jitesh Dugar: Do you remember, at the time of loading the data, we need an embedding model here? So now, Anetan is asking us, tell me the embedding model that I should use.

[02:06:50] - [02:06:51]

Jitesh Dugar: So?

[02:06:51] - [02:07:02]

Jitesh Dugar: These are all the possible embedding models that you can use, and because there is OpenAI, OpenAI also provides an embedding model, apart from a chat model, so we will select embeddings.

[02:07:03] - [02:07:08]

Jitesh Dugar: The credentials will remain the same, and you can choose a model, here.

[02:07:09] - [02:07:12]

Jitesh Dugar: And... You can close it.

[02:07:12] - [02:07:17]

Jitesh Dugar: But before that, you should go to Option, and choose Dimensions.

[02:07:18] - [02:07:22]

Jitesh Dugar: Do you remember what dimensions we had used for the vector store?

[02:07:22] - [02:07:24]

Jitesh Dugar: 1024.

[02:07:24] - [02:07:29]

Jitesh Dugar: We have to make sure that the embedding model also uses the same dimension.

[02:07:30] - [02:07:40]

Jitesh Dugar: Because if embedding model is breaking down into maybe 300 dimensions, but the vector store is capable of storing 1,000 dimensions, it will give you a mismatch. It will not work.

[02:07:40] - [02:07:47]

Jitesh Dugar: So, embedding models should also break in the same number of dimension that the vector database expects.

[02:07:48] - [02:07:53]

Jitesh Dugar: That's why we will select dimensions here, and we will select 1024.

[02:07:54] - [02:07:56]

Jitesh Dugar: And with that, we will close it.

[02:07:58] - [02:08:01]

Jitesh Dugar: So, embeddings model is chosen. The next option is

[02:08:02] - [02:08:07]

Jitesh Dugar: document. Now, it is asking, tell me the document that you want to load into the vector store.

[02:08:07] - [02:08:12]

Jitesh Dugar: Click on the plus button, and the option... the only option you have is Default DATA Loader.

[02:08:12] - [02:08:13]

Jitesh Dugar: Select that.

[02:08:14] - [02:08:20]

Jitesh Dugar: And under this, you have an option that says JSON. Change that to binary.

[02:08:20] - [02:08:23]

Jitesh Dugar: Remember, there is JSON and binary.

[02:08:23] - [02:08:28]

Jitesh Dugar: In the world of automations, A file is called a binary.

[02:08:28] - [02:08:30]

Jitesh Dugar: So, you should select binary here.

[02:08:31] - [02:08:35]

Jitesh Dugar: Mode is load all input data, all that is okay, and...

[02:08:36] - [02:08:41]

Jitesh Dugar: Text splitting is simple. Is it okay to keep it to simple text splitting, everyone?

[02:08:42] - [02:08:44]

Jitesh Dugar: Or should we choose something else?

[02:08:52] - [02:08:54]

Jitesh Dugar: Yes, something else? Yes, that's right.

[02:08:59] - [02:09:03]

Jitesh Dugar: Yes, we should use something else, and some of you answered it correctly also.

[02:09:03] - [02:09:06]

Jitesh Dugar: We should use custom for now, and we'll see what options we get after that.

[02:09:07] - [02:09:23]

Jitesh Dugar: And by the way, I will just repeat, over here we selected binary because JSON means it's a line of JSON text. We are not using a JSON text, we are using a file. File in the workflow space is called a binary. So we select binary.

[02:09:23] - [02:09:34]

Jitesh Dugar: And text splitting will be custom, and we close it, and the moment we do that... by the way, let me... this is a part of the same workflow canvas, though, but it's separated slightly.

[02:09:37] - [02:09:37]

Jitesh Dugar: Yeah.

[02:09:41] - [02:09:55]

Jitesh Dugar: Now, the moment we select binary, sorry, custom text splitting, it shows up an extra space here, extra option to choose what text splitter we want to use, and you guys know the answer to it. You know that we have to use a

[02:09:56] - [02:10:00]

Jitesh Dugar: Character text splitter, or a recursive character text splitter, or the others.

[02:10:04] - [02:10:12]

Jitesh Dugar: Yes, the recursive character text printer, that's correct. We choose that, and it is asking us for the chunk size and the chunk overlap.

[02:10:12] - [02:10:16]

Jitesh Dugar: Do you remember what is a recommended chunk size to start with?

[02:10:19] - [02:10:21]

Jitesh Dugar: Yeah, okay, great.

[02:10:21] - [02:10:26]

Jitesh Dugar: 500 recommended chunk size. If you do not remember that, this cheat sheet already has it.

[02:10:26] - [02:10:37]

Jitesh Dugar: Start with a 500 character text chunk size, and use a 50 to 100 character chunk overlap. So, let's start with a 100 character text chunk overlap.

[02:10:38] - [02:10:41]

Jitesh Dugar: Now, with that, this is all done.

[02:10:41] - [02:10:42]

Jitesh Dugar: And?

[02:10:43] - [02:10:47]

Jitesh Dugar: We do not have to get into the process of doing all the chunky and everything, we can simply run this.

[02:10:48] - [02:10:54]

Jitesh Dugar: And it will automatically do all of the data ingestion steps for us, everything that we did in Phase 1.

[02:10:54] - [02:10:56]

Jitesh Dugar: Everything will be handled.

[02:10:57] - [02:11:01]

Jitesh Dugar: What is chunk overlap? Chunk overlap was the process where there was

[02:11:02] - [02:11:06]

Jitesh Dugar: A common text between the first chunk and the second chunk.

[02:11:07] - [02:11:13]

Jitesh Dugar: Now, with that, this has run successfully. If we go back to our vector database, and if I try to refresh it.

[02:11:16] - [02:11:21]

Jitesh Dugar: You can already see there are all of these chunks added inside our vector database.

[02:11:21] - [02:11:27]

Jitesh Dugar: everything is added here. You can try to read the text version of it also, so you can see what each chunk includes.

[02:11:28] - [02:11:33]

Jitesh Dugar: But this has been stored inside your vector database. So, our database is now ready.

[02:11:34] - [02:11:36]

Jitesh Dugar: Everyone good until this point.

[02:11:38] - [02:11:42]

Jitesh Dugar: I will be doing a recap towards the end again. I will be doing a recap towards the end.

[02:11:43] - [02:11:54]

Jitesh Dugar: But the magic of all of this is, while I explained all of this in possibly about 45 minutes, you could implement this on anton because you had to simply configure these steps.

[02:11:55] - [02:12:07]

Jitesh Dugar: You had to simply configure what should be the chunk size, what should be the chunk overlap, all of these you are able to do. Now, I got a question. Why can't we directly load it here? If you try to directly load it, it will not create those chunks.

[02:12:08] - [02:12:13]

Jitesh Dugar: And do not worry about being able to recall. If you practice it once on your own, you will be able to recall this.

[02:12:17] - [02:12:24]

Jitesh Dugar: So... This step that you saw here, this was our data ingestion step. So this whole thing.

[02:12:24] - [02:12:27]

Jitesh Dugar: I've just added a note here.

[02:12:28] - [02:12:37]

Jitesh Dugar: And I would just say... This is our data ingestion step of our... RAC pipeline. Step 1.

[02:12:38] - [02:12:40]

Jitesh Dugar: Which is data ingestion.

[02:12:40] - [02:12:42]

Jitesh Dugar: That's what happens here.

[02:12:42] - [02:12:47]

Jitesh Dugar: So we have now loaded our document into the vector database. That's all we've done.

[02:12:49] - [02:12:53]

Jitesh Dugar: Next part is getting answers to the questions.

[02:12:54] - [02:13:03]

Jitesh Dugar: getting... using that vector database to get answers to the questions. That's what we are supposed to do. And that's something that we will do in this step here.

[02:13:04] - [02:13:14]

Jitesh Dugar: That's something that we'll do in this step, where we have to give our AI agent an ability to use the vector store to find answers to a question.

[02:13:15] - [02:13:21]

Jitesh Dugar: Now, so for the AI agent, please pay attention what I'm doing now. We will click on the tool button.

[02:13:22] - [02:13:26]

Jitesh Dugar: There is a built-in There is a built-in

[02:13:27] - [02:13:35]

Jitesh Dugar: tool within NATEN that handles all of the question-answering phase for you, the entire RAG pipeline, it handles all of it.

[02:13:36] - [02:13:45]

Jitesh Dugar: This entire thing will be handled by AI... by a tool in NATEN, and that is called Vector Store Question Answer Tool.

[02:13:46] - [02:13:50]

Jitesh Dugar: As you scroll down, there's something called Vector Store Question Answer Tool.

[02:13:50] - [02:13:52]

Jitesh Dugar: So you can select that.

[02:13:53] - [02:13:58]

Jitesh Dugar: And you can give it a name, you can say, This is my... Big house.

[02:13:59] - [02:14:01]

Jitesh Dugar: Knowledge base.

[02:14:03] - [02:14:09]

Jitesh Dugar: And then you can give it a description, because this description will be used by the AI agent. You can just say.

[02:14:09] - [02:14:15]

Jitesh Dugar: Use this tool to get answers to, questions on Big House customer support policy.

[02:14:23] - [02:14:26]

Jitesh Dugar: So, this is a description that you're giving to the tool.

[02:14:28] - [02:14:33]

Jitesh Dugar: And now... Now... This thing is now asking us.

[02:14:34] - [02:14:46]

Jitesh Dugar: Tell me the vector store that I should use. Tell me the vector store that I should use. By the way, what is the limit for? That is the limit of how many results it should return, and then the AI agent can figure out the right answer also.

[02:14:48] - [02:14:51]

Jitesh Dugar: Vector store is what it is asking now, and

[02:14:52] - [02:15:04]

Jitesh Dugar: You remember what vector store we were using? Out of all of these, it was Pinecone. So we will select Pinecone, and we will tell it, this is the location, or this is the vector store that you're supposed to get answers from.

[02:15:04] - [02:15:07]

Jitesh Dugar: So, we will come here, and we will select.

[02:15:07] - [02:15:11]

Jitesh Dugar: Bakehouse Knowledge Base, the one that we loaded in Step 1.

[02:15:13] - [02:15:21]

Jitesh Dugar: So, we select that, and we can close it. So, vector store is something that is now already connected to this knowledge base.

[02:15:21] - [02:15:30]

Jitesh Dugar: And it is also asking us the embedding model. We already used an embedding model previously, and we used OpenAI's embedding model, we will use that.

[02:15:31] - [02:15:38]

Jitesh Dugar: Let's use text embedding 3 itself, and dimension we have to choose 1024, because that's the one that we are using across.

[02:15:38] - [02:15:44]

Jitesh Dugar: So now, this also needs an embedding model, because it needs a way to convert the question into a vector.

[02:15:45] - [02:15:54]

Jitesh Dugar: And finally, it also is asking for a chat model. Click on this plus button, and let's use OpenAI chat model again.

[02:15:57] - [02:15:58]

Jitesh Dugar: And?

[02:16:00] - [02:16:02]

Jitesh Dugar: One minute, I want to take a pause right now.

[02:16:02] - [02:16:07]

Jitesh Dugar: Do you think I'm going too fast? I thought I was just selecting a few things here, but...

[02:16:10] - [02:16:16]

Jitesh Dugar: Because I'm literally, I'm literally just selecting... okay, I will, I will just slow down, okay?

[02:16:17] - [02:16:18]

Jitesh Dugar: Okay.

[02:16:18] - [02:16:25]

Jitesh Dugar: Everyone, all that I've done is, I first added a vector store question-answer tool, Okay?

[02:16:25] - [02:16:28]

Jitesh Dugar: And after that, it is giving us two options.

[02:16:28] - [02:16:35]

Jitesh Dugar: Where it is now asking us to choose the vector store, click on the plus button, And then...

[02:16:35] - [02:16:41]

Jitesh Dugar: We will scroll down, because we are using Pinecone, where our vector store

[02:16:41] - [02:16:43]

Jitesh Dugar: is added. So, we will select

[02:16:44] - [02:16:47]

Jitesh Dugar: Pinecone Vector Store, where we have added the vector store.

[02:16:47] - [02:16:51]

Jitesh Dugar: And our Pinecone credentials are already connected.

[02:16:51] - [02:16:58]

Jitesh Dugar: And we will select the Pinecone Index. We used Bakehouse Weck Knowledge Base, where we loaded the

[02:16:59] - [02:17:01]

Jitesh Dugar: Document, so we will select that.

[02:17:02] - [02:17:07]

Jitesh Dugar: And we will close it. So we are now telling this, that you need to refer to this vector database.

[02:17:09] - [02:17:13]

Jitesh Dugar: Now it is asking us, What is the embedding model that we can use?

[02:17:14] - [02:17:15]

Jitesh Dugar: Click on the plus button.

[02:17:16] - [02:17:21]

Jitesh Dugar: And... search, look for embeddings, which is OpenAI embeddings.

[02:17:22] - [02:17:27]

Jitesh Dugar: Select OpenAI embeddings, Our OpenAI credentials are already connected.

[02:17:28] - [02:17:34]

Jitesh Dugar: Model, you can use it as text embedding 3 small itself, and then you need to specify the dimensions.

[02:17:35] - [02:17:39]

Jitesh Dugar: And you all know what dimension we had used. It was 1024.

[02:17:40] - [02:17:50]

Jitesh Dugar: That's the dimension that we used for the original vector store, that's the one we used for the training also, the data ingestion also, and we will use the same thing to get the answers as well.

[02:17:53] - [02:17:57]

Jitesh Dugar: So, you close it, and with that, the embedding model is also connected.

[02:17:57] - [02:18:02]

Jitesh Dugar: And now, this also needs access to a chat model, an LLM.

[02:18:03] - [02:18:06]

Jitesh Dugar: So... Click on the plus button.

[02:18:06] - [02:18:10]

Jitesh Dugar: And, again, use the OpenAI chat model.

[02:18:10] - [02:18:16]

Jitesh Dugar: And our credentials are connected, you can use GPT-5 Pini and simply close it, that's it.

[02:18:16] - [02:18:19]

Jitesh Dugar: Now, if you're wondering, Everyone.

[02:18:20] - [02:18:36]

Jitesh Dugar: I will repeat this for you, and I'll also answer a few questions. Now, if you're wondering, why did you add the chat model again here when the AI agent already has a chat model? Okay? And why did you add this embeddings OpenAI again? Why do you need this? Now, please pay attention.

[02:18:37] - [02:18:39]

Jitesh Dugar: So, in the question-answering phase.

[02:18:39] - [02:18:46]

Jitesh Dugar: Do you remember, this question-answering phase has one embedding model, and it has one large language model?

[02:18:47] - [02:18:53]

Jitesh Dugar: So, the job of this chat model is just simply to help with the rack pipeline.

[02:18:53] - [02:19:00]

Jitesh Dugar: It will get a question, So, the way this thing works, the way this whole thing works, right.

[02:19:01] - [02:19:07]

Jitesh Dugar: This whole thing, everything that I have highlighted right now, this is our entire rack pipeline.

[02:19:07] - [02:19:17]

Jitesh Dugar: This is our entire rack pipeline. It will get a question from the AI agent. What this AI... the rack pipeline will do is, it will go to that vector store.

[02:19:18] - [02:19:22]

Jitesh Dugar: Okay? And before that, for that question, it will first convert that into

[02:19:22] - [02:19:29]

Jitesh Dugar: a query vector using this embedding model. It will go to that vector store, find out the relevant chunks.

[02:19:30] - [02:19:33]

Jitesh Dugar: Find out the relevant chunks, And then...

[02:19:33] - [02:19:43]

Jitesh Dugar: It will augment that with the question that it got, it will feed that to this chat model, it will get an answer, and it will give that answer back to the AI agent.

[02:19:45] - [02:19:47]

Jitesh Dugar: Did I make sense, everyone?

[02:19:48] - [02:19:54]

Jitesh Dugar: On the rack pipeline aspect, why do we need two embeddings models and two chat models, rather?

[02:19:55] - [02:20:05]

Jitesh Dugar: I will explain it one more time, okay? Please pay attention, okay? So, the job of this part, the job of this part.

[02:20:05] - [02:20:10]

Jitesh Dugar: is... to take a question and give you an answer. That's all.

[02:20:11] - [02:20:20]

Jitesh Dugar: That's all, okay? And how does it work? The AI agent will send a question to this. It may give you a question like, hey, what is the refund policy in this situation?

[02:20:21] - [02:20:30]

Jitesh Dugar: What this does, it automatically uses the embedding model to convert that into a vector, a query vector, exactly what we explained in the first place here.

[02:20:31] - [02:20:33]

Jitesh Dugar: Then, It...

[02:20:33] - [02:20:40]

Jitesh Dugar: Goes to the vector store that it is already connected to. It will find all the relevant chunks that is closest to the question.

[02:20:41] - [02:20:43]

Jitesh Dugar: That is likely to have an answer.

[02:20:43] - [02:20:46]

Jitesh Dugar: Then, it will send the question?

[02:20:46] - [02:20:54]

Jitesh Dugar: in the form of vectors, along with the relevant chunks, it will augment both of these together, and it will give it to this LLM model.

[02:20:56] - [02:20:57]

Jitesh Dugar: And...

[02:20:57] - [02:21:04]

Jitesh Dugar: After that, it will generate an answer in plain English, and this tool will give back that answer to the AI agent.

[02:21:05] - [02:21:10]

Jitesh Dugar: People who are not clear, are you clear on the RAG pipeline concept now?

[02:21:12] - [02:21:22]

Jitesh Dugar: Remember, technical implementation, how to connect things here and there, these are things that you can anytime try. Understanding stuff is what is a bit challenging.

[02:21:27] - [02:21:29]

Jitesh Dugar: On a scale of 1 to 10,

[02:21:30] - [02:21:33]

Jitesh Dugar: Are you clear right now? On a scale of 1 to 10, how clear are you?

[02:21:34] - [02:21:38]

Jitesh Dugar: 10787865, okay? 7, 8, 9, 10, okay?

[02:21:40] - [02:21:41]

Jitesh Dugar: Okay, perfect.

[02:21:42] - [02:21:43]

Jitesh Dugar: Perfect, okay.

[02:21:45] - [02:21:54]

Jitesh Dugar: need to practice. I really recommend you to practice this, because as I said, you are leveling up. Every day, we have to level up. We cannot talk about very simple stuff.

[02:21:55] - [02:21:55]

Jitesh Dugar: Right?

[02:21:56] - [02:22:03]

Jitesh Dugar: Now, with that, we have this added the vector sort question-answer tool, and if we come back to our whiteboard.

[02:22:03] - [02:22:11]

Jitesh Dugar: We also need to give our AI agent an ability to reply to our email. So, we'll connect it to another tool.

[02:22:13] - [02:22:14]

Jitesh Dugar: Which will be Gmail.

[02:22:14] - [02:22:16]

Jitesh Dugar: Click on the plus button.

[02:22:17] - [02:22:19]

Jitesh Dugar: Search for Gmail.

[02:22:20] - [02:22:26]

Jitesh Dugar: Select Gmail tool. Do not select human in the Loop Gmail, that's for a different purpose. Select Gmail tool.

[02:22:27] - [02:22:34]

Jitesh Dugar: And... After that, Message send is what we have to choose here.

[02:22:34] - [02:22:43]

Jitesh Dugar: Let me see if there's a way to reply. Yeah, there's a way to reply also. Do you all agree, then, that rather than sending a new message, we should reply to an existing email?

[02:22:44] - [02:22:49]

Jitesh Dugar: We should reply to an existing email, rather than sending a new message. So...

[02:22:49] - [02:22:53]

Jitesh Dugar: The customer support email that we got, that's what we're supposed to reply to.

[02:22:54] - [02:22:57]

Jitesh Dugar: So, I will choose operation as reply.

[02:22:57] - [02:23:02]

Jitesh Dugar: After that, it will ask us the message ID that you want to reply to.

[02:23:02] - [02:23:04]

Jitesh Dugar: Where will we get the message ID, everyone?

[02:23:06] - [02:23:07]

Jitesh Dugar: Where do we get the message ID?

[02:23:10] - [02:23:22]

Jitesh Dugar: Yes, on the left-hand side on the Gmail trigger, that's correct. So, expand the Gmail trigger on the left-hand side. The message ID is right here. This is the ID of the email that we are supposed to reply to. I will copy that, and I will put it here.

[02:23:23] - [02:23:23]

Jitesh Dugar: Next.

[02:23:24] - [02:23:25]

Jitesh Dugar: Next.

[02:23:26] - [02:23:32]

Jitesh Dugar: The content of the email Okay, what is...

[02:23:32] - [02:23:34]

Jitesh Dugar: Okay, some of you are saying I made a mistake.

[02:23:37] - [02:23:39]

Jitesh Dugar: Okay, I made a mistake. What is that mistake?

[02:23:40] - [02:23:48]

Jitesh Dugar: Yeah, make it dynamic, yes, that's correct. So instead of putting a fixed message ID value here, this ID should be dynamic, you guys are correct.

[02:23:48] - [02:23:53]

Jitesh Dugar: So, I have to make this dynamic, which means I'll have to drag it, and I'll have to put it here.

[02:23:54] - [02:23:57]

Jitesh Dugar: Which means it'll reply to that email now.

[02:23:58] - [02:23:59]

Jitesh Dugar: Next.

[02:23:59] - [02:24:01]

Jitesh Dugar: the message content.

[02:24:02] - [02:24:04]

Jitesh Dugar: the message content.

[02:24:04] - [02:24:07]

Jitesh Dugar: Where will the message content come from?

[02:24:08] - [02:24:12]

Jitesh Dugar: Where will the message content come from? Snippet? No, you guys are wrong.

[02:24:13] - [02:24:19]

Jitesh Dugar: It will not be the snippet, because we are supposed to reply with a message. Snippet is the email that we got.

[02:24:19] - [02:24:22]

Jitesh Dugar: We are supplied... supposed to reply to the message.

[02:24:22] - [02:24:30]

Jitesh Dugar: So, the message, we don't know what the message will be, but who will know what the message will be? The AI agent.

[02:24:30] - [02:24:45]

Jitesh Dugar: So we need to give our AI agent an ability to control what to put in this field. And for that, there is a button here that says, let the model define this parameter. I like to call it the Magic AI button, so you can click on the Magic AI button, which means...

[02:24:45] - [02:24:51]

Jitesh Dugar: The moment you select this, AI will magically fill up the reply message content.

[02:24:53] - [02:24:54]

Jitesh Dugar: And with that, you are done.

[02:24:55] - [02:25:04]

Jitesh Dugar: And if you would like to add anything else, let's say you want to remove NA10's branding, you can select that. Let me not do that for now, and we'll... I'll show you later.

[02:25:05] - [02:25:14]

Jitesh Dugar: With that, AI Agent now has an ability to reply to an email, it has an ability to look up the knowledge base as well, using a full-blown RAC pipeline.

[02:25:15] - [02:25:18]

Jitesh Dugar: Now... Memory, I don't think it's as important.

[02:25:19] - [02:25:37]

Jitesh Dugar: as important, but let me still add a memory. Now, when you try to add a memory, click on the memory button, and it gives you an option to select simple memory. I would recommend use a simple memory that NA10 provides. If you want to use external memories, that's up to you, but start with a simple memory.

[02:25:38] - [02:25:42]

Jitesh Dugar: And then... Simple Memory is now asking for the session ID.

[02:25:43] - [02:25:47]

Jitesh Dugar: session ID. Now, the AI agent needs to remember the conversations.

[02:25:48] - [02:25:52]

Jitesh Dugar: AI agent needs to remember the conversation. So, let me pull up

[02:25:53] - [02:25:57]

Jitesh Dugar: Something where I want to ask all of you a question.

[02:25:57] - [02:25:59]

Jitesh Dugar: Let's say this is your AI agent.

[02:26:00] - [02:26:04]

Jitesh Dugar: This AI agent simultaneously might be talking to hundreds of customers.

[02:26:06] - [02:26:12]

Jitesh Dugar: it might be talking to possibly 3 people. Okay, I'm putting a few names from chat.

[02:26:12] - [02:26:13]

Jitesh Dugar: ROBERT?

[02:26:14] - [02:26:20]

Jitesh Dugar: let's say Janani, Okay, and let's say J.

[02:26:22] - [02:26:23]

Jitesh Dugar: So...

[02:26:24] - [02:26:35]

Jitesh Dugar: If... if the AI agent is talking to these three people, do you all agree that it needs some kind of a unique way to store the conversations for all 3 of them separately?

[02:26:35] - [02:26:41]

Jitesh Dugar: It should never mix up the messages of Janani with Roberts, or Jay with Roberts.

[02:26:42] - [02:26:51]

Jitesh Dugar: Yes, everyone agrees to that. So, it needs one unique identifier for every user that it is talking to. Can name be that unique identifier?

[02:26:51] - [02:26:53]

Jitesh Dugar: Can we store based on the name?

[02:26:54] - [02:26:59]

Jitesh Dugar: We cannot, because multiple people can have the same name. We can use something more unique.

[02:26:59] - [02:27:02]

Jitesh Dugar: And what is that unique identifier in our case?

[02:27:02] - [02:27:10]

Jitesh Dugar: It will be the... thread ID, the email thread ID, not the message ID, the email thread ID.

[02:27:10] - [02:27:15]

Jitesh Dugar: Because it needs to treat each of the threads of the email separately.

[02:27:16] - [02:27:21]

Jitesh Dugar: So we will select... I will define the session ID below, and select the thread ID from here.

[02:27:22] - [02:27:30]

Jitesh Dugar: Now, people who are asking, why not message ID? Because every time you reply to the message, the message ID will change.

[02:27:30] - [02:27:32]

Jitesh Dugar: I will show you an example.

[02:27:42] - [02:27:44]

Jitesh Dugar: Let me pull up one example.

[02:27:50] - [02:27:56]

Jitesh Dugar: Okay? Let me pull up a random message, okay? So... This is one random message.

[02:27:57] - [02:28:03]

Jitesh Dugar: So... What happens here is... In this case, this is a very test random message.

[02:28:03] - [02:28:22]

Jitesh Dugar: This is an email thread conversation. So, there was an email sent by this person, someone replied to it, someone else replied to it, someone else replied to it. Now, the thing with all of those four email messages, the message ID will be different for every single email, but across these threads, the thread ID will remain the same.

[02:28:23] - [02:28:29]

Jitesh Dugar: The thread ID will remain the same, and that's the reason we should use thread ID as that unique identifier.

[02:28:31] - [02:28:36]

Jitesh Dugar: So, if someone replies to the mail, the message ID will change, but thread ID will remain the same.

[02:28:38] - [02:28:39]

Jitesh Dugar: In case...

[02:28:39] - [02:28:47]

Jitesh Dugar: In case you are confused, you can also decide to use the email address. That also will work fine in majority of the cases.

[02:28:48] - [02:28:52]

Jitesh Dugar: With that, Your memory is also added.

[02:28:52] - [02:28:55]

Jitesh Dugar: Is there something that we have still missed, everyone?

[02:28:55] - [02:28:57]

Jitesh Dugar: Is there something that we've still missed?

[02:28:58] - [02:29:02]

Jitesh Dugar: We will run this step. Can we run the AI agent, or have we missed something? We've added the...

[02:29:03] - [02:29:09]

Jitesh Dugar: Brain, we've added the memory, we've added the tools also. Two tools are added. Did we miss something?

[02:29:17] - [02:29:20]

Jitesh Dugar: We have forgotten to add the system prompt.

[02:29:21] - [02:29:27]

Jitesh Dugar: We have forgotten to add the system prompt. So, double-click on the AI agent, and I'll show you how to add the system prompt.

[02:29:28] - [02:29:33]

Jitesh Dugar: You open the AI agent, Under Option, click on Add Option.

[02:29:33] - [02:29:36]

Jitesh Dugar: Select System Message, and you have to give a system prompt here.

[02:29:37] - [02:29:37]

Jitesh Dugar: So?

[02:29:38] - [02:29:41]

Jitesh Dugar: The system prompt is where you tell it how it should act.

[02:29:42] - [02:29:45]

Jitesh Dugar: Let me use Claude to generate a system prompt quickly.

[02:29:47] - [02:29:51]

Jitesh Dugar: I need your help to generate a system prompt for my...

[02:29:51] - [02:29:58]

Jitesh Dugar: customer support AI agent for Bakehouse, bakery.

[02:29:58] - [02:30:12]

Jitesh Dugar: And, you have access to two tools. One is a Gmail reply tool, and the other one is a Bakehouse Customer Support Knowledge Base. You need to look at the incoming email and help me reply to the incoming email.

[02:30:12] - [02:30:27]

Jitesh Dugar: By looking at the knowledge base, make sure to always refer to the knowledge base to get the most accurate responses. The responses should be human-like and polite, and you can use a signature that ends with Kelly from Bakehouse.

[02:30:32] - [02:30:37]

Jitesh Dugar: So, we have asked Klaus to generate a good system prompt for us.

[02:30:37] - [02:30:40]

Jitesh Dugar: And let me also give it one more information.

[02:30:42] - [02:30:45]

Jitesh Dugar: Though... The...

[02:30:45] - [02:30:54]

Jitesh Dugar: Gmail reply tool allows you to use HTML as the email reply, so make sure to format it in a good way, using HTML message.

[02:30:57] - [02:31:01]

Jitesh Dugar: Why are we using HTML? Because HTML emails are slightly better formatted.

[02:31:17] - [02:31:26]

Jitesh Dugar: Okay, you are Kelly, a customer support agent for Bakehouse Bakery. You reply to customer emails on behalf of the bakery. Your tools are these. Gmail reply.

[02:31:26] - [02:31:45]

Jitesh Dugar: How to handle every email, read the email, search the knowledge base, write the reply only using what the knowledge base returned. Accuracy rules, these are not optional. Tone, warm, polite, send valid email, use the structure, right? Use P tags only, no div, and all of these. So, we can copy this, and we will put it in the system prompt.

[02:31:45] - [02:31:48]

Jitesh Dugar: We used Klaus to generate the system prompt.

[02:31:48] - [02:31:51]

Jitesh Dugar: Everything will be shared with you. And with that, we can run this.

[02:31:52] - [02:31:55]

Jitesh Dugar: Step... It is using the brain, it's using the memory.

[02:31:56] - [02:31:58]

Jitesh Dugar: And it should use the knowledge base also.

[02:32:00] - [02:32:04]

Jitesh Dugar: And it should use the Gmail reply tool. It is using the knowledge base to find the answers.

[02:32:10] - [02:32:15]

Jitesh Dugar: It is... It augmented, and it generates... it is using the LLM to generate the answer.

[02:32:17] - [02:32:19]

Jitesh Dugar: Let's see if we get a reply shortly.

[02:32:32] - [02:32:37]

Jitesh Dugar: There you go, it replied to the email. Let me open up my mailbox and see if I got a reply.

[02:32:47] - [02:33:03]

Jitesh Dugar: Yes, there you go. It said, Hi Jitesh, I'm really sorry about your delivery was delayed by more than 2 hours. That's a frustrating experience. We'd like to look into this with the delivery team. If you can reply with your order number and the time it was delivered, we'll check what happened and follow up. And this is what it replied with.

[02:33:04] - [02:33:06]

Jitesh Dugar: Do you think it is good, everyone?

[02:33:10] - [02:33:18]

Jitesh Dugar: I think it did a really good job. Of course, it can be made better, but it's still a good one, considering we did not spend a lot of effort on building the prompt.

[02:33:18] - [02:33:24]

Jitesh Dugar: And... And a lot of these things you can improve by improving the prompt also.

[02:33:24] - [02:33:33]

Jitesh Dugar: And everything happens automatically. It looked up the knowledge base. And if you're curious on the RAG stuff, let me open the... this step here, and I want to show you something.

[02:33:33] - [02:33:39]

Jitesh Dugar: On the left-hand side, the AI agent asked a question. This is not a question that we asked.

[02:33:39] - [02:33:55]

Jitesh Dugar: to the knowledge base. The AI agent asked a question, what is Bakehouse policy for delayed delivery complaints? Is there a compensation refund? And it replied something also. It said the policy text does not give any automatic compensation, specifically for deliveries that are simply delayed by more than 2 hours.

[02:33:56] - [02:33:58]

Jitesh Dugar: So that's what we are mentioning. So, the...

[02:33:59] - [02:34:06]

Jitesh Dugar: RAC pipeline looked up the knowledge base, and it said, this is what happens, and AI agent decides what to reply to it.

[02:34:08] - [02:34:14]

Jitesh Dugar: And with that, You can publish this workflow, and we will test it out, everything, automatically.

[02:34:15] - [02:34:18]

Jitesh Dugar: So, you have published this workflow, And...

[02:34:19] - [02:34:28]

Jitesh Dugar: Once this workflow is published, you can send an email. We can send an email to our inbox again, and we expect to see an email coming back automatically.

[02:34:28] - [02:34:31]

Jitesh Dugar: So let me... Send a new email.

[02:34:32] - [02:34:35]

Jitesh Dugar: To the same email address again.

[02:34:36] - [02:34:41]

Jitesh Dugar: So, I would say... To this same email address.

[02:34:42] - [02:34:47]

Jitesh Dugar: I would say, I would possibly ask a question specifically around this to make sure it's answering correctly.

[02:34:47] - [02:34:48]

Jitesh Dugar: Okay.

[02:34:48] - [02:34:50]

Jitesh Dugar: What are your store timings, and how do I order?

[02:34:53] - [02:34:58]

Jitesh Dugar: Hi, Bakehouse team, what are your store timings, and how do I place an order? What are the different ways?

[02:35:01] - [02:35:04]

Jitesh Dugar: And I would go ahead and send it. And once this happens.

[02:35:05] - [02:35:12]

Jitesh Dugar: I'm not doing anything at all. We don't have to run all of these steps at all. You can just go to executions and watch for it to run.

[02:35:13] - [02:35:15]

Jitesh Dugar: And remember, the poll time that we set was 1 minute.

[02:35:16] - [02:35:17]

Jitesh Dugar: So?

[02:35:18] - [02:35:23]

Jitesh Dugar: After 1 minute, Within a minute, you should see an execution coming here.

[02:35:26] - [02:35:40]

Jitesh Dugar: Now, if you're wondering, by the way, while it is waiting for the email, I also wanted to show you, we use the same workflow canvas to create two separate workflows, but they are both independent. Both of them have a separate trigger.

[02:35:40] - [02:35:52]

Jitesh Dugar: You can decide to put these in two separate workflows also, that's up to you. But whenever you get a new email, it never runs this. This will only run whenever you want to prepare that knowledge base.

[02:35:53] - [02:35:56]

Jitesh Dugar: Whenever your knowledge base updates, only then this workflow will run.

[02:35:56] - [02:36:03]

Jitesh Dugar: This workflow will run whenever someone sends you a message, and as you can see, it is running.

[02:36:04] - [02:36:06]

Jitesh Dugar: The execution is happening right now.

[02:36:09] - [02:36:12]

Jitesh Dugar: Maybe the email also came. Email has come in.

[02:36:14] - [02:36:18]

Jitesh Dugar: It's still running right now. 20 seconds, 21 seconds so far.

[02:36:19] - [02:36:23]

Jitesh Dugar: There you go, it has replied in 18 seconds, let's see if we have a reply.

[02:36:27] - [02:36:28]

Jitesh Dugar: Oh, wait.

[02:36:31] - [02:36:33]

Jitesh Dugar: Where there has been an issue.

[02:36:33] - [02:36:35]

Jitesh Dugar: Where it did not send a reply.

[02:36:38] - [02:36:40]

Jitesh Dugar: Wait, there has been a small issue.

[02:36:41] - [02:36:44]

Jitesh Dugar: It did not send a reply, we have to be a little more explicit on the prompt.

[02:36:46] - [02:36:47]

Jitesh Dugar: One minute.

[02:36:56] - [02:37:01]

Jitesh Dugar: you should... Always reply to every email.

[02:37:01] - [02:37:06]

Jitesh Dugar: So this is one more step that we are adding, and we will have it processed one more time.

[02:37:06] - [02:37:09]

Jitesh Dugar: And we will have it processed again for the same email.

[02:37:17] - [02:37:19]

Jitesh Dugar: Yeah, I think it sent a reply this time.

[02:37:19] - [02:37:26]

Jitesh Dugar: So I think the issue was, the prompt did not specify that they have to always send a reply. And with that, we have a reply.

[02:37:26] - [02:37:43]

Jitesh Dugar: And the reply looks like this. I don't have the store opening and closing hours, so I can't confirm those to you right now. However, you can place an order on the website by WhatsApp, phone, or in person at the store. Delivery starts a 2-hour window between 9am to 9pm. Let's look at the document here.

[02:37:47] - [02:37:53]

Jitesh Dugar: Delivery... how do you place an order? There you go. You can place an order through website, WhatsApp.

[02:37:53] - [02:37:55]

Jitesh Dugar: Phone and walk-in.

[02:37:55] - [02:37:58]

Jitesh Dugar: And that's the information that this also includes.

[02:37:59] - [02:38:02]

Jitesh Dugar: And delivery slots are 2-hour windows.

[02:38:02] - [02:38:07]

Jitesh Dugar: whether the 2R windows are mentioned, right? So, let me search for it.

[02:38:09] - [02:38:13]

Jitesh Dugar: There you go, you... there is a delivery slot of 2 hours window.

[02:38:14] - [02:38:17]

Jitesh Dugar: Between 9 AM to 9pm, that's... is what is mentioned here.

[02:38:18] - [02:38:22]

Jitesh Dugar: Now, another question, why did it miss the store opening hours?

[02:38:24] - [02:38:26]

Jitesh Dugar: One minute, let me check.

[02:38:30] - [02:38:33]

Jitesh Dugar: I think we will change the prompt slightly again, and we will say...

[02:38:39] - [02:38:43]

Jitesh Dugar: Always double-check on the knowledge base.

[02:38:48] - [02:38:49]

Jitesh Dugar: Before answering.

[02:38:52] - [02:38:54]

Jitesh Dugar: And we will publish it one more time.

[02:38:54] - [02:38:58]

Jitesh Dugar: And we'll try another message again. We'll try one more message.

[02:38:58] - [02:39:02]

Jitesh Dugar: And this time, we will again ask something related to

[02:39:04] - [02:39:07]

Jitesh Dugar: the refund policy. We'll send another message.

[02:39:07] - [02:39:11]

Jitesh Dugar: And we'll say... What is your refund policy?

[02:39:14] - [02:39:17]

Jitesh Dugar: Can you tell me about your refund policy?

[02:39:22] - [02:39:25]

Jitesh Dugar: And we'll go to Executions.

[02:39:27] - [02:39:31]

Jitesh Dugar: Oh, actually, we can ask the store opening times. Let's pro- send multiple emails also.

[02:39:35] - [02:39:37]

Jitesh Dugar: What are your store opening hours?

[02:39:39] - [02:39:41]

Jitesh Dugar: Tell me about your store opening hours, please.

[02:39:43] - [02:39:45]

Jitesh Dugar: So, we're sending it two emails.

[02:39:45] - [02:39:46]

Jitesh Dugar: Both of them?

[02:39:54] - [02:40:02]

Jitesh Dugar: And let's see if we've gotten the emails here. Yeah, I think the first email has landed, and we are waiting for it. It is processing the workflow right now.

[02:40:02] - [02:40:04]

Jitesh Dugar: All the executions will happen.

[02:40:23] - [02:40:26]

Jitesh Dugar: Yes, it has sent a reply, let's see what the reply looks like.

[02:40:27] - [02:40:28]

Jitesh Dugar: The first one?

[02:40:31] - [02:40:42]

Jitesh Dugar: If you cancel before the delivery store, there's no charge, and you get a full refund. Bakehouse cancels the order, you get a full refund plus a 15% discount. Let's see if all that is mentioned here.

[02:40:44] - [02:40:49]

Jitesh Dugar: Yeah, if Bakehouse cancels the refund, you get a 15% discount code.

[02:40:49] - [02:40:51]

Jitesh Dugar: That's mentioned here. Let's look at the next one.

[02:40:53] - [02:40:54]

Jitesh Dugar: Store opening hours.

[02:41:00] - [02:41:05]

Jitesh Dugar: The second one hasn't been processed, so we'll wait for the second execution to happen.

[02:41:07] - [02:41:09]

Jitesh Dugar: Yeah, the second one is running right now.

[02:41:12] - [02:41:16]

Jitesh Dugar: Yeah, it processed the second one also, and let us run it.

[02:41:17] - [02:41:18]

Jitesh Dugar: Let us see what happened.

[02:41:20] - [02:41:23]

Jitesh Dugar: Yeah, this is the store opening hours. It replied, but...

[02:41:23] - [02:41:28]

Jitesh Dugar: And it can process 10 emails at once, so it is able to send us a reply, and of course.

[02:41:28] - [02:41:39]

Jitesh Dugar: One of the things that it did was, when it... when we asked it multiple questions, it wasn't able to understand. What we could do is, we could improve the model. Remember, we are using a mini model right now.

[02:41:39] - [02:41:45]

Jitesh Dugar: If we see such issues where it is not able to answer properly, switch to a better model.

[02:41:46] - [02:41:48]

Jitesh Dugar: And that's how it can do a better job also.

[02:41:49] - [02:41:58]

Jitesh Dugar: Now, with that, we saw a few examples where AI is automatically replying to all the emails that we are sending, all by looking up their knowledge base. Now.

[02:41:58] - [02:42:15]

Jitesh Dugar: With that, our workflow is completed. I will send you all this workflow. You can download it, or you can import that on your site. Go ahead and practice it. And I highly recommend all of you to practice it. Just to do a recap of the entire workflow that we have.

[02:42:15] - [02:42:20]

Jitesh Dugar: So... And this part that you see in yellow.

[02:42:20] - [02:42:25]

Jitesh Dugar: This part is the data ingestion step, and you do it only once.

[02:42:25] - [02:42:28]

Jitesh Dugar: You do the data ingestion part only once.

[02:42:28] - [02:42:33]

Jitesh Dugar: Whenever you have a new document, and you load that inside this vector store.

[02:42:35] - [02:42:38]

Jitesh Dugar: Once it is loaded inside that vector store.

[02:42:38] - [02:42:47]

Jitesh Dugar: This step here, this tool here, only consumes that vector store to give answers to the AI agent for the question that it is asking.

[02:42:49] - [02:42:50]

Jitesh Dugar: So?

[02:42:51] - [02:42:55]

Jitesh Dugar: And of course, you might also wonder why did we use the classify step, so that

[02:42:55] - [02:43:02]

Jitesh Dugar: AI Agent, we are not wasting its resources for all the questions that are not related to customer support.

[02:43:03] - [02:43:11]

Jitesh Dugar: Creating that vector database is only a one-time activity. Only if your knowledge base or the customer support policy changes, you'll have to run it again.

[02:43:11] - [02:43:14]

Jitesh Dugar: That's what happens. How do you update it?

[02:43:16] - [02:43:20]

Jitesh Dugar: You can come here, and you can load it one more time until insert again.

[02:43:20] - [02:43:24]

Jitesh Dugar: But, there is another option here that says Update Documents.

[02:43:25] - [02:43:28]

Jitesh Dugar: You can use it to update the documents as well.

[02:43:31] - [02:43:35]

Jitesh Dugar: Now, with that, this is our entire pipeline. You can clean this up.

[02:43:36] - [02:43:39]

Jitesh Dugar: And let me add some sticky notes here as well.

[02:43:51] - [02:43:54]

Jitesh Dugar: Let's call it... Mrs.

[02:43:55] - [02:43:56]

Jitesh Dugar: Step 2.

[02:43:57] - [02:43:59]

Jitesh Dugar: Answers... Questions?

[02:44:00] - [02:44:01]

Jitesh Dugar: Using RAG.

[02:44:03] - [02:44:09]

Jitesh Dugar: This is the other one, which we will just change the color to something like this.

[02:44:09] - [02:44:13]

Jitesh Dugar: So these are the two parts of our entire rack pipeline.

[02:44:15] - [02:44:15]

Jitesh Dugar: No.

[02:44:16] - [02:44:23]

Jitesh Dugar: on a scale of 1 to 10, I know this was tough, okay? This was slightly advanced. On a scale of 1 to 10,

[02:44:23] - [02:44:28]

Jitesh Dugar: How confident are you Are you in terms of being executing it yourself?

[02:44:29] - [02:44:34]

Jitesh Dugar: 10, 8, 7, 6.5, 7.5, 10.7, okay.

[02:44:34] - [02:44:36]

Jitesh Dugar: Okay, I can see a lot of you are saying more than 5.

[02:44:37] - [02:44:43]

Jitesh Dugar: Which itself is a huge win. And let me be honest, RAG is not an easy concept, and we are just leveling up.

[02:44:43] - [02:44:45]

Jitesh Dugar: And congratulations if you're beyond 5.

[02:44:45] - [02:44:54]

Jitesh Dugar: And I encourage everyone to go ahead and try it. We will have a workbook exercise also after this on a customer support use case.

[02:44:55] - [02:44:58]

Jitesh Dugar: With that, everyone, thank you everyone for being so patient throughout.

[02:44:59] - [02:45:03]

Jitesh Dugar: And, I absolutely enjoyed building this workflow alongside you.

[02:45:05] - [02:45:06]

Jitesh Dugar: Over to you, Shashti.

[02:45:08] - [02:45:08]

Shasti: Yep.

[02:45:09] - [02:45:32]

Shasti: Thank you so much, Jitesh. I think that was a wonderful session. I can see the hearts and reactions popping up on my screen, and also the chat. How was it, folks? There is a poll that is launched right in front of you. Go ahead and give your ratings as well. As Jitesh mentioned, yes, it was a complex topic, and I think Jitesh did a wonderful job from breaking down the complex topic of rags.

[02:45:32] - [02:45:46]

Shasti: to building a full-fledged customer support agent using Inn. So, if you ever felt like this was too intense, again, we have the solution for it. You will have the access to recordings. Go ahead, watch it maybe once or twice.

[02:45:46] - [02:46:01]

Shasti: And then you will be able to grab it, and you'll be able to build one for yourself as well. All the JSON files, the whiteboard screenshots, everything will be sent to you on your LMS as part of resources post the session. So, I hope that is clear.

[02:46:03] - [02:46:12]

Shasti: And I see a lot of you have not yet given your feedback on the poll, so do take 2 minutes and let us know how this session went for you, right?

[02:46:13] - [02:46:27]

Shasti: Yes, we do have the workbook, and we will be getting into it, but before that, just wanted to understand if you guys need a quick 5 minutes break before we get into the break, and then we will come back and discuss on the breakout room workbook.

[02:46:27] - [02:46:34]

Shasti: In the meantime, team, we can also drop the workbook link, reflection link, and the webinar ID, and then we get into break.

[02:46:39] - [02:46:57]

Shasti: Okay, I see a lot of requests for 10 minutes and 5 minutes. Okay, quick check, folks, do we need a 10 minutes break or 5 minutes break? Again, disclaimer, if you're taking 10 minutes break, we might have to extend our breakout rooms by 10 minutes. Only if you're not gonna complain about extending our breakout rooms, I would be okay giving a 10 minutes break.

[02:46:59] - [02:47:02]

Shasti: Let me know if you have a deal, then I'm happy giving a 10 minutes break.

[02:47:05] - [02:47:16]

Shasti: Okay, great. Perfect. So, team, whenever we are ready, we can quickly take a 10 minutes break, and we come back and talk about what we are going to do in the breakout rooms, and we'll get into the breakout room activity.

[02:55:52] - [02:55:54]

Audio shared by Sumedha: da-da-da-da-da-da.

[02:57:28] - [02:57:37]

Shasti: Okay, everyone, are we guys back from the break? Drop a back in the chat if you guys are back from the break. I hope the 10 minutes was helpful.

[02:57:40] - [02:57:41]

Shasti: I'm back.

[02:57:42] - [02:57:56]

Shasti: Great. Perfect. Now that we all are back, without any further ado, we'll quickly get started. So I'll quickly share my screen and give you a walkthrough of what exactly are we going to do in today's breakout room, and then we will get into the breakout room.

[02:57:58] - [02:58:02]

Shasti: Drop a quick yes in the chat if my screen is visible.

[02:58:05] - [02:58:06]

Shasti: Yes.

[02:58:09] - [02:58:10]

Shasti: Great.

[02:58:10] - [02:58:32]

Shasti: So this is the workbook that we are going to solve inside the breakout room today, right? The use case is the same. You are going to build a customer support Rayage bot in the breakout

rooms using NATN, but the workflow is not same as what we learned just now, because what we learned is not something that we can do inside the breakout rooms in a span of 1.5 to 2 hours.

[02:58:32] - [02:58:57]

Shasti: Because to set up your rack system and vector database, it takes a lot of time. It also depends on your system, right? That's why what we have done is, instead of giving you the same workflow, we have given you a workbook where you'll be building customer support TriEdgebot, which basically, this is how the workflow will look like, which will take the feedback or a query from a customer, and AI agent will read it, understand if that needs an escalation.

[02:58:57] - [02:59:21]

Shasti: Or if it is... defines the urgency of the situation, and if it is a high urgency bug, it... what it's gonna do is, it will be sending an email back to the customer, and it will also escalate the issue to the team, saying that so-and-so issue has been raised by so-and-so customer, and you need to look into it, and also sends an acknowledgement email to the customer who has written the ticket.

[02:59:21] - [02:59:36]

Shasti: This is what we are going to do inside the breakout room today, right? In today's session, inside the breakout rooms, we are going to use the chat as a trigger, but post-session, if you are somebody who actually want to go ahead and build it for your Gmail.

[02:59:37] - [02:59:52]

Shasti: We have also given a tab which says optional, which you can go ahead and use Gmail as your trigger after the session is done. But during the breakout rooms, we will focus on chat trigger option, because that is more feasible and easy for most of you who are very new to this.

[02:59:52] - [02:59:57]

Shasti: Right? That is the objective of the breakout room as well, for you to get an understanding of how this

[02:59:57] - [03:00:04]

Shasti: We can make use of the agent and automate the stuff that would require a lot of manual, time and manual efforts.

[03:00:04] - [03:00:13]

Shasti: Post-session, you can go ahead and use Gmail as a trigger for the same workflow, where customer raises the ticket on your Gmail, and agent will do its job.

[03:00:13] - [03:00:26]

Shasti: This is what we are going to do in today's breakout room. Are we clear, folks? For that, whatever you need, the step-by-step process, the API key, everything, and our amazing commenters will be there to help you.

[03:00:26] - [03:00:33]

Shasti: Now, questions about API key? Same as yesterday, this API key will remain active till tomorrow, 6pm IST.

[03:00:34] - [03:00:46]

Shasti: So, if you're somebody who's not able to participate in today's breakout room activity, you have time till tomorrow, 6pm IST, to go ahead and do the workbook by your own. Post that, the API key will be revoked from our end.

[03:00:46] - [03:00:48]

Shasti: I hope that is clear.

[03:00:49] - [03:01:03]

Shasti: Yes, we did anticipate questions about, can we also have workbook 6pm IST is when the API key will be deactivated, so please make sure you're using it before 6pm IST tomorrow.

[03:01:04] - [03:01:15]

Shasti: Amarja, no, the API key of yesterday's workbook was active till today, 6pm, IST, so it was working perfectly fine.

[03:01:20] - [03:01:32]

Shasti: Great. Now, the questions about can we get the workbook for the same workflow that we did, which is, you know, building your vector database and adding your RAG concept. So, what we have done is, if you go to your LMS,

[03:01:33] - [03:01:38]

Shasti: And scroll down to day... Four...

[03:01:39] - [03:01:47]

Shasti: And just click on Resources section. Folks, the questions about API key, yes, you can go ahead and create your own API keys as well.

[03:01:51] - [03:02:06]

Shasti: If you go ahead on LMS and click on Day 4, you have something called Additional Workbook, right? Just click on this Additional Workbook, and you will have the workbook for the same workflow that Jitesh has explained today.

[03:02:07] - [03:02:26]

Shasti: So, whoever wants to go ahead and do the advanced workbook, please do it post-session, in session, we will be following the customer support Trayj bot, and all the resources, the whiteboard, the JSON file, everything that was used by Jitesh will be shared post-session on your LMS under the same resources section.

[03:02:27] - [03:02:32]

Shasti: Please do go and search under Day 4 Resources, you will see the additional workbook.

[03:02:33] - [03:02:34]

Shasti: Hope that is clear.

[03:02:38] - [03:02:55]

Shasti: Great. With that said, we are pretty much good to get started with our breakout rooms, and we have all the amazing commenters from yesterday who will be guiding you and helping you in the breakout rooms. We will have the breakout rooms for the next 1.5 to 2 hours.

[03:02:55] - [03:03:02]

Shasti: Right? And we would want all of you to go ahead and build your first ever customer support Triad bot. I know

[03:03:02] - [03:03:18]

Shasti: Most of you might think that it is not that easy. That's why we have this breakout room activities planned for you, where you get help from co-mentors. So please do go ahead, give it a shot, even if you are not doing it hands-on, see how it works, and then maybe you might get the confidence to do it on your own.

[03:03:18] - [03:03:34]

Shasti: And I also saw a couple of questions about yesterday's workbook, that we learned the same thing in the first 3 hours, and then we built the same thing in the breakout room. What if we would want an additional use case? Folks, that just says that you are not yet exploring LMS to the fullest.

[03:03:34] - [03:03:44]

Shasti: So go ahead on Day 3 Resources, you have an additional workbook where the use case is you building your expense tracking system on an ATM.

[03:03:44] - [03:04:01]

Shasti: Right? That's a whole different use case from what we learned yesterday. So go ahead and explore that as well if you do have time. We did provide you an additional use case that you can go ahead and try on, but it's just that you never went ahead and explored LMS. So I'm assuming that tomorrow when you are coming back, you will not have this question.

[03:04:01] - [03:04:06]

Shasti: Somebody says, what does LMS stand for? It's Learning Management System.

[03:04:07] - [03:04:23]

Shasti: Great. With that said, I hope all your questions about the workbook and resources are cleared, and team, whenever we are ready, we can drop the workbook and webinar ID in the chat for one last time, and we can get started with the breakout rooms.

[03:04:23] - [03:04:30]

Shasti: Breakout rooms are not yet created, folks. Please do have a bit of patience. Once it is created, you will have the pop-up on your screen.

[03:04:31] - [03:04:35]

Shasti: The expense tracking workbook is on Day 3 LMS, folks.

[03:04:45] - [03:04:59]

Shasti: There you go, the breakout rooms are open. You will get a pop-up on your screen if you're joining from Zoom app. If you're joining from Zoom web, I cannot help you guys, so please do drop off and join from Zoom app, and you can join your breakout rooms.

[03:05:33] - [03:05:51]

Shasti: Okay, I see a couple of you saying that you're not able to join the breakout rooms. Let me try to send an invite again for you guys, and I can only do that if you have joined from Zoom app and not Zoom web. Rahul, I've sent you the invite. Please do accept the pop-up on your screen.

[03:05:57] - [03:06:02]

Shasti: Ranjith, you have also got an invite to the breakout room. Please do accept.

[03:06:08] - [03:06:17]

Shasti: Yes, folks, every resource that the mentor wanted to share with you, which includes the whiteboard and the JSON file, will be uploaded on LMS under your resources.

[03:06:20] - [03:06:32]

Shasti: I do not see expense tracking. Mahesh, if you're still here, it is available under Day 3 Resources section. Please do go ahead and explore.

[03:07:17] - [03:07:23]

Shasti: Sorry, my bad, I was on mute. I'm going through the questions that are there on the chat, I'm responding to each of you.

[03:07:23] - [03:07:25]

Shasti: Please do send.

[03:07:25] - [03:07:41]

Shasti: For somebody who was asked that, send the link to join the breakout rooms, there is no link to join the breakout rooms. You will get a pop-up on your screen to join the breakout rooms. All you need to do is click on join, if you're joining from Zoom app. If you're not joining from Zoom app, you cannot join the breakout rooms.

[03:07:42] - [03:07:47]

Shasti: Ranjith, send the request again for you to join the breakout rooms. Please do accept.

[03:07:49] - [03:07:53]

Shasti: Sneha, I'm sending the request again.

[03:07:53] - [03:07:55]

Shasti: Please do accept the pop-up.

[03:07:59] - [03:08:17]

Shasti: Mahesh, great. If you have already completed the workbook, you are one step above all of us. Do... maybe you can still join the breakout rooms and maybe help the other folks, other peers who are trying to join, along with a mentor. That would be really appreciated, but if not, you can stay back as well.

[03:08:20] - [03:08:30]

Shasti: Okay, Mahesh, let me quickly check and help you with where exactly you can find the expense tracker workbook. Give me a second.

[03:08:42] - [03:08:47]

Shasti: Yeah, Gilbert, I have sent an invite again to join Breakout Room 1. Please do accept it.

[03:09:56] - [03:10:15]

Shasti: folks, whoever is facing issue joining the breakout room, again, to repeat, if you're on Zoom web, you will not be able to join the breakout rooms, and there is no link that I can share. As soon as you join from a Zoom app, you will get an invite, on your screen itself to join a breakout room.

[03:10:18] - [03:10:26]

Shasti: Mahesh, our team is looking into it. Allow us a couple of seconds, it should reflect on your LMS in the next 2-3 minutes. I'll let you know once that is done.

[03:10:31] - [03:10:44]

Shasti: Ghulam, sorry if I'm pronouncing your name wrong, but I can see that you have already got an invite to join a breakout room. I'm sending the request again. You'll get a pop-up on your screen, please do accept.

[03:10:47] - [03:10:50]

Shasti: Yeah, you should have gotten the invite.

[03:10:58] - [03:11:11]

Shasti: I still see a lot of you here in the main room, folks. Please do drop it in the chat if you're facing any issues joining the breakout room, I'll be able to help. And if you are opting to stay back here and solve the workbook on your own, that is also great.

[03:44:39] - [03:44:51]

Shasti: Hey folks, whoever is done with the workbook, I'm gonna drop the reflection link and the feedback form link in the chat. Go ahead, generate your badges, and also drop the feedback on how your breakout rooms went.

[03:44:52] - [03:44:54]

Shasti: And then you guys can drop.

[03:44:55] - [03:45:08]

Shasti: Yeah, if you're done with the reflections already, thank you so much for joining in today's session, and I hope you built your first ever customer support traged bot in the breakout rooms. Go ahead, drop the workflow screenshot, or let us know how it went.

[03:45:08] - [03:45:18]

Shasti: for you in the gratitude channel, we would love to see it, and we'll see you guys again tomorrow at 6pm IST, if you are joining the Bonus Office Hours, 7.30 for the main session. Thank you so much.

[03:56:04] - [03:56:22]

Shasti: And if you were not able to get it cleared in the breakout room, one is you can post it on the Ask channel on your LMS, and the mentor will be, getting back to you with a response in the next 24 hours. Second, what I would recommend is you can join the bonus office hours that we have planned for you, which will happen at 6pm IST tomorrow.

[03:56:23] - [03:56:27]

Shasti: And you can ask in that bonus office hours as well. I hope that is clear.

[03:56:29] - [03:56:30]

Shasti: Yeah.

[03:56:32] - [03:56:44]

Shasti: Folks, again, whoever is done with the workbook, if you have already generated your reflection link for the day and given your feedback, go ahead and drop off and get good amount of rest. We will be seeing you guys tomorrow.

[03:58:11] - [03:58:23]

Shasti: Okay. Ajay, the AI Companion option, you will see a small icon on your right-hand side of your Zoom screen. That is what we, keep saying as the AI Companion.

[03:58:24] - [03:58:25]

Shasti: And,

[03:58:26] - [03:58:44]

Shasti: Folks, if you want the reflection link of previous days, please do go ahead and ask on the tech support channel, just specify which day you would want, and there is an AIAP bot that will give you the links that you would want instantly, but make sure you're asking on the tech support channel with the day that you would want the reflection link for.

[03:58:48] - [03:59:04]

Shasti: If you're asking how to generate the batch, whenever we drop the reflection link here in the chat, just open it. All you need to do is enter your name, and give the 3 key takeaways that

you have learned today, and just click on generate batch, that's it you will have to do, and it will generate the batch.

[03:59:07] - [03:59:19]

Shasti: Ajay says he completed the workbook today. Great, Ajay, go ahead and post it on the LMS. We would love to see the outcome that you have driven in the breakout rooms, and I'm assuming that you have already filled the feedback form as well.

[03:59:21] - [03:59:27]

Shasti: Yes, we have shared the reflection link and the feedback form, sharing it again on the chat for all of you folks.

[04:05:49] - [04:05:57]

Shasti: Yes, folks, if you're done with the workbook and generated your reflection and filled the feedback form, please do go ahead and drop off. We'll be seeing you tomorrow.

[04:18:31] - [04:18:43]

Shasti: Folks, the Day 3 Additional Workbook is under Resources section. It is named as Additional Workbook, so go ahead and check out. Let me see if I can drop the

[04:18:43] - [04:18:49]

Shasti: link here, which I'm not supposed to, because you all should go ahead and explore LMS.

[04:18:50] - [04:18:54]

Shasti: But since most of you are facing issue, let me see if I can fetch the link.

[04:18:58] - [04:18:58]

Shasti: Yeah.

[04:19:12] - [04:19:17]

Shasti: I've dropped the additional workbook off... Day 3, so...

[04:19:17] - [04:19:19]

Shasti: Go ahead and try it out.

[04:19:23] - [04:19:30]

Shasti: Roohi, yes, I will have the team have a check on this. I've noted your email as well.

[04:19:30] - [04:19:35]

Shasti: So, do not worry about that. You should be getting a response very soon.

[04:19:37] - [04:19:40]

Shasti: If you folks are done, go ahead, drop off.

[04:19:40] - [04:19:41]

Shasti: We'll see you guys tomorrow.

[04:29:03] - [04:29:16]

Shasti: Hey folks, if you're done with the workbook, and if you're back here in the main room, go ahead and drop off, and I'm assuming that you'd have filled the feedback form of your breakout rooms, and also generated your reflection batch.

[04:29:16] - [04:29:24]

Shasti: So, we'll be seeing you guys tomorrow at 6pm IST, if you're planning to join the Bonus Office Hours, or 7.30pm

[04:29:24] - [04:29:28]

Shasti: ISD, if you want to join the main session, and make sure you're join...

[04:29:29] - [04:29:38]

Shasti: Running live to the main session, because we will be doing a deep dive on advanced MCPs tomorrow, so I do not want you guys to miss that particular session, because it's gonna be fun.

[04:29:39] - [04:29:40]

Shasti: Thank you so much.

[04:29:50] - [04:30:08]

Shasti: Whoever is asking for a particular day batch link, go ahead on your LMS. There is a channel called Tech Support. Drop your query over there, you will get the link instantly, because we have a bot who's looking into those tickets. So, please do ask for a particular day, and you will get it over there.

[04:39:37] - [04:39:37]

Karam Hawash_Outskill: Kind of...

[04:39:41] - [04:39:55]

Shasti: Okay, everyone, we are back in the main room. How was it, folks? How many of you were able to build your first-ever customer support RajBot in your breakout rooms? Drop a me in the chat if you have successfully done it.

[04:39:58] - [04:40:03]

Shasti: Okay, I see those reactions on my screen and also the chat.

[04:40:04] - [04:40:17]

Shasti: It was great. API key did not work for you. That should not be the case, because it works for all of us. Maybe if you have copied it with a space or something, that might be the issue. Try it again, by copying the exact API key.

[04:40:18] - [04:40:41]

Shasti: Team, we can also drop the breakout room feedback link in the chat. Folks, I'm seeing most of you have successfully done the breakout room activity and did the workbook, but are not taking a minute to give us the feedback form, because that would help us, you know, work on what we can improve for our upcoming breakout rooms and also the future cohorts. So, it will take a minute of yours.

[04:40:41] - [04:40:48]

Shasti: So please do go ahead and drop your feedback on how it went for all of you, and we would love to take a look.

[04:40:48] - [04:40:51]

Shasti: The reflection link is also there.

[04:40:51] - [04:41:12]

Shasti: How do I post my work to Outskill? If you're referring about the, agent that you built, the workflow, take a screenshot of the workflow, go ahead and post it on the reflection channel. Also, all the amazing commenters, thank you so much for joining today and helping our learners. With that, I'll let you go and get some good amount of rest. We will be seeing you soon.

[04:41:14] - [04:41:20]

Shasti: Yes, whoever is done with the feedback form and reflection badge, go ahead and drop off.

[04:41:20] - [04:41:37]

Shasti: Also, folks, I'm not seeing many of you interacting on LMS. That is a place where all of you should go ahead and talk to the community, interact with the community, show your gratitude for all the amazing commenters and the team who's helping you all. So go ahead and post it on the LMS, we would love to take a look.

[04:41:37] - [04:41:40]

Shasti: Let's be a supportive community, right?

[04:41:40] - [04:41:57]

Shasti: I'll wait for another 2 minutes for those of you who have not yet filled the feedback form, and for those of you who are dropping, tomorrow we'll connect back again at 6pm IST for our bonus office Hours and 7.30pm IST, where we are going to do a deep dive on advanced MCPs.

[04:41:58] - [04:42:17]

Shasti: We have got glimpses of MCP and what is MCP, what are connectors. On day one of our journey, if you guys do remember, we learned on what are connectors, how do we use it on ChatGPT and Clot, but tomorrow we are going to do a deep dive on how you can create your own MCP server, and it is going to be fun if you're...

[04:42:17] - [04:42:36]

Shasti: or somebody would like to experiment new things and new stuff. So make sure, even if you're not able to make it to the office hours, make it live to the 7.30 session, and I'm sure you're gonna love it. And we'll also discuss about how Day 6 and Day 7 is going to look like, if you are somebody who do not have a clarity yet.

[04:42:36] - [04:42:40]

Shasti: Right? I hope that is clear on how our tomorrow's agenda is going to look like.

[04:42:43] - [04:42:51]

Shasti: Where can we see the Q&A transcript? So, Q&A transcript, is available on post-read section of your LMS.

[04:42:51] - [04:42:59]

Shasti: And you can go ahead and explore it on day 1, or day 2, or Day 3 on your LMS. You will find it under post streets of each day.

[04:43:03] - [04:43:12]

Shasti: Great. With that, once again, thank you so much, everybody, for joining your Day 4 of the program, and we are almost at the end of

[04:43:12] - [04:43:30]

Shasti: week 1 of our program, we only have 3 days left, so make sure you're making it live for the rest of the 3 days to make the most out of it. And, kudos to all of you for staying up to date and building your first ever agent. And do post on LMS, we will be connecting again tomorrow at 6pm.

[04:43:30] - [04:43:33]

Shasti: Thank you so much, everybody. Have a great night.